

CAN VISION-LANGUAGE MODELS LEARN TO EXECUTE CLASSICAL SEARCH ALGORITHMS?

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ABSTRACT

Whether vision-language models learn classical search depends on measuring choice. Random-valid, uniform over valid operations, makes the reference’s decisions on 48/48 additive pairs of our enumeration contract, holding every expanded-baseline process-SFT success. A tie-break change gives 19/48 divergent pairs, though a pre-registered prediction failed. Validating before scoring, our choice-frontier contract passes the identity audit and counts decisions, and M1 passes a pre-registered privileged ladder on validation and held-out panels, validating the measurement, not a policy.

A trained scene-only adapter chooses frontier states in greedy best-first and weighted A* better than random-valid ($D_3 = +0.285 [+0.140, +0.433]$), without approaching the reference. No enumeration-contract BFS or BFWS cell shows learned success. An unscreened panel gives $+0.149 [+0.030, +0.283]$ from 4 of 12 tasks (descriptive). Separation above the rung randomizing $\epsilon=0.75$ of reference decisions holds only on the held-out panel ($+0.227 [+0.053, +0.403]$), not on the validation or unscreened panels or pooled. ScienceWorld is a positive control (0.915 versus 0). On LLM-First Search, random-valid solves every task under token caps by construction, but 0.0775 at the reference’s expansions. The 48/48 failure appears only on our runtime, held-out means one 11-task panel, the frozen 45-task final evaluation is unexecuted, and claims are development-stage, with no planning-ability or generality claim.

1 INTRODUCTION

To ask whether a vision-language model can learn to execute a classical search algorithm, we need a success measure that responds to the model’s choices. Prior probes evaluate language models on classical planning (Valmeekam et al., 2023b; Stein et al., 2025; Kokel et al., 2026). We instead train a Search Process Policy to emit Typed Search Operations, which a no-repair Trusted Search Runtime validates (Section 3). The declared algorithms are breadth-first search (BFS), Best-First Width Search (BFWS), and additive best-first search, which is best-first search ordered by the additive heuristic h_{add} . The policy acts under one of two decision contracts, with text, visual, or multimodal observations under the first and scene-only observations under the second. The natural control is random-valid, which chooses uniformly over valid operations.

Random-valid can make every decision the reference makes, and then a learned gain is validity, not choice. Under our enumeration contract, each expansion submits its complete candidate set. In its additive best-first cells, random-valid makes exactly the decisions of the exact reference, the declared algorithm run as specified, on 48/48 task-algorithm pairs (Figure 1a). Process supervised fine-tuning (SFT) solved 125 of the expanded baseline’s 288 task-algorithm-observation entries, all in these cells, and 0/24 in each BFS and BFWS cell (Section 5.1). In the greedy additive replication cell on the same 24-task panel, SFT beats the base at every seed ($+0.875$ at two of three) and never beats random-valid (-0.125 at those two, descriptive). The replication uses seeds 17/29/71 and is descriptive, with no pre-registered rule (Statistics appendix, Table 5).

The identity is a property of executed code, which the contract specification does not reveal. It follows from the runtime’s serial rule, which breaks ties in sorted candidate order (Counterfactual appendix, Appendix S). The identity audit, a decision-by-decision comparison of random-valid with the exact reference, finds 19/48 divergent pairs when ties instead break in submission order. A pre-

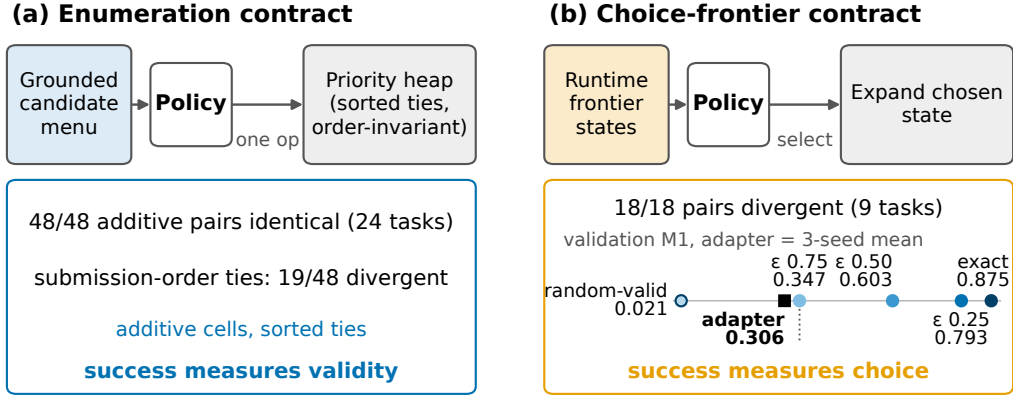


Figure 1: Validate before score (steps 1 and 3 of Box 1). Random-valid success measures choice only on an interface where its decisions can diverge from the exact reference’s, and the measure must order privileged reference-derived selectors before it scores any policy. (a) Under the enumeration contract, random-valid makes the exact reference’s decisions on 48/48 additive pairs, and 19/48 pairs diverge when ties break in submission order. (b) Under the choice-frontier contract, 18/18 pairs on the reused 9-task panel diverge. On the validation panel, M1 falls from 0.875 for the exact reference, a value fixed by construction, through the exact- ϵ rungs, which randomize a fraction ϵ of reference decisions, to 0.021 for random-valid (point estimates, with intervals in Figure 2). The dotted line marks the 0.75 rung. The adapter’s three-seed mean is 0.306. Success therefore measures validity in the additive cells of (a) under sorted tie-breaking and choice in (b), and the rungs read privileged information. In the BFS and BFWS cells of (a), random-valid’s decisions diverge and its success falls below the reference’s, and submission-order ties make 19/48 pairs divergent (Section 5.1).

registered prediction of more than 24/48 divergent pairs failed. Saturation alone does not separate the cases, because random-valid also saturated the BFS development panel. On the expanded baseline, random-valid’s BFS decisions nonetheless diverge from the reference’s, and it solves 15/24 against 24/24 (Section 5.1).

We therefore validate the measurement before scoring any policy, starting from a choice-frontier contract in which the policy picks the runtime-owned frontier state to expand. It passes the identity audit with 18/18 divergent pairs (Figure 1b), and its budget counts decisions, twice the reference’s expansion count, applying the compute-matched-baseline principle (Li & Talwalkar, 2020). M1, the area under the solve-versus-budget curve, must then pass a pre-registered ladder of exact- ϵ selectors, which randomize a fraction ϵ of reference decisions. On the validation panel, 12 fresh tasks from 7 domains, the ladder passes with smallest adjacent gap $+0.082$ [$+0.051$, $+0.111$]. It passes again on the held-out panel, with smallest gap $+0.130$ [$+0.070$, $+0.190$] (Section 5.2). The rungs read the reference’s computed priority order, which is never shown to a policy, so the ladder validates the measurement, not a policy, and Box 1 states the procedure.

The validated measurement scores the zero-shot base and an earlier small-corpus adapter at the random-valid level and the trained adapter above it. The zero-shot base is equivalent to random-valid on the validation panel (development stage), and the earlier small-corpus adapter scores at its level (descriptive, Section 5.3). The trained adapter’s gain D_3 , its M1 minus random-valid’s averaged over three training seeds, is $+0.285$ [$+0.140$, $+0.433$] on the validation panel, with every seed positive. On the held-out panel, frozen before any evaluation episode, $D_3 = +0.416$ [$+0.250$, $+0.591$]. On the unscreened panel, of tasks random-valid never solved in screening, the gain is $+0.149$ [$+0.030$, $+0.283$], resting on 4 of 12 tasks (descriptive). The adapter is separated above the 0.75 rung only on the held-out panel, not on the validation panel, the unscreened panel, or pooled (Table 1), and the adapter is trained only by imitating the exact reference.

Steps 1 and 2 of Box 1 run on two published interfaces, one audited instance each. ScienceWorld (Wang et al., 2022) is a positive control, with 1000/1000 divergent pairs over 200 tasks and success of 0.915 versus 0. No audited-candidate interface uses an enumeration contract, so the 48/48 failure is shown only on our runtime. LLM-First Search (Herr et al., 2025) is a by-construction illustration

of the budget step, since a zero-token chooser solves every task under its token caps. Counted in expansions, random-valid solves 0.0775 at the reference’s expansion count (Section 5.4).

This paper makes three contributions, the first primary.

- **A validate-before-score search-choice measurement, admitted by the identity audit and a decision-counted budget (the compute-matched-baseline principle of Li & Talwalkar, 2020), whose M1 passes a pre-registered privileged ladder on validation and held-out panels before any policy is scored.**
- **Evidence that the admission step is needed. On one runtime the identity audit returns both verdicts from a tie-break change alone, and it locates every process-SFT success on the expanded baseline in cells where success measures validity.**
- **A worked use and a portable procedure (Box 1). The measurement scores the trained adapter above random-valid on three panels, and the zero-shot base and an earlier small-corpus adapter at its level. Its first two steps run on two published interfaces, one audited instance each, crediting the budget step to the compute-matched-baseline principle (Li & Talwalkar, 2020).**

Every claim is development-stage, measured on development panels, not on the frozen 45-task final evaluation, which is unexecuted. Held-out means only the fresh 11-task held-out panel, drawn from seeds no adapter had been evaluated on (Section 4). Enumeration-contract cells are single-seed except two replication cells (seeds 17/29/71, descriptive $n=3$). We make no planning-ability or generality claim.

2 RELATED WORK

We evaluate step-by-step execution of declared algorithms such as A* (Hart et al., 1968) or BFWS (Lipovetzky & Geffner, 2017), not plans. Planning benchmarks (Valmeekam et al., 2023a;b; Stein et al., 2025; Kokel et al., 2026) score plans or plan judgments, and off-the-shelf language models improve at operational reasoning but not at structural enumeration (Huang et al., 2026). Trace-trained models (Lehnert et al., 2024; Gandhi et al., 2024) and step imitators (Veličković et al., 2020; Zaremba & Sutskever, 2014) likewise lack typed operations and in-episode validation. Our choice-frontier contract, where the model chooses which frontier state to expand, is a learned node-selection policy (Orseau & Lelis, 2021). The submission-order counterfactual isolates a heap tie-breaking effect of the kind studied for A* search (Asai & Fukunaga, 2016).

Our model issues the operations that determine exploration, whereas in Tree of Thoughts (Yao et al., 2023) and Reasoning via Planning (RAP) (Hao et al., 2023) a scaffold or upper-confidence policy owns them. Scratchpads (Nye et al., 2021), per-step labels (Lightman et al., 2024), Visual Sketchpad (Hu et al., 2024), and Multimodal Visualization-of-Thought (MVoT) (Li et al., 2025) supervise or generate unchecked free-form steps and imagery. Unlike repair loops (Kambhampati et al., 2024; Madaan et al., 2023) and syntax-only constrained decoding (Scholak et al., 2021; Willard & Louf, 2023), our runtime never repairs and checks algorithmic invariants on every operation.

Shortcut work locates the confound in the model or the data, and ours lives in the runtime. Evaluation-validity work warns that common-sense tasks can confound planning with retrieval (Valmeekam et al., 2023a), and embodied suites extend evaluation coverage (Li et al., 2024). Multimodal-training research documents visual question answering (VQA) models relying on language priors rather than image content (Zhu et al., 2020), and video-language evaluation reports answer-position selection bias (Loginova et al., 2025). Label-position selection bias is documented for LLM multiple-choice selectors (Zheng et al., 2024), and large vision-language model (LVLM) chain-of-thought audits find visual or format cues can influence answers without appearing in reasoning traces (Balasubramanian et al., 2025). Our choice-frontier contract instead permutes candidate labels per decision, the standard mitigation rather than a contribution. Process supervision (Lightman et al., 2024) supplies intermediate-step feedback but does not test whether a grounded action menu makes frontier choices identifiable.

Matched-budget random baselines and valid-action handicaps are known, and we apply them. Random search at matched compute is the standard architecture-search baseline (Li & Talwalkar, 2020), and relative-to-random reporting removes an expertly engineered search space’s advantage (Yang

et al., 2020), so we apply this matched-baseline principle as a budget check and add no new budget principle. Jericho and the Contextual Action Language Model (CALM) treat valid-action access as a disclosed handicap (Hausknecht et al., 2020; Yao et al., 2020), and no per-decision test checks whether uniform choice already reproduces the reference. ScienceWorld (Wang et al., 2022) exposes per-step choice among valid actions and is a positive control, since random-valid registers choice there. This paper operationalizes the known admissible-action handicap in search-execution evaluation with a per-decision identity audit that returns both verdicts on one runtime and admits the validated choice measurement of Section 5. We also validate the success measure against selectors of known order before scoring any policy, and the rungs read privileged information, so this checks the measurement, not a policy.

3 SEARCH PROCESS POLICY

A Search Process Policy executes a declared search algorithm one typed operation at a time, and the runtime makes every one of its decisions checkable. An episode fixes (x, A, m, B) , a task x in a deterministic grounded planning domain, a declared algorithm A given as input, the observation adapter m , distinct from the trained adapter scored in Section 5, and a frozen budget B . At step t , m renders the decision context as a text, visual, or multimodal observation o_t that exposes every fact the exact reference uses. Under the enumeration contract, Search Memory (the runtime-held search state defined below) and the candidates stay textual.

The runtime owns all search state and every check, and the policy owns only the operations it emits. **Typed Search Operation.** The policy emits a type from A ’s vocabulary plus every exploration-determining operand, and an emission missing one is invalid. **Search Memory.** This is bounded runtime state (frontier, visited table, BFWS novelty counters), not model state, and o_t is the policy’s only channel to history.

Algorithm Invariant. The runtime checks deterministic per-algorithm properties on every operation. BFS must expand in first-in, first-out (FIFO) order, and BFWS must follow novelty and goal-count priority (Lipovetzky & Geffner, 2017). The two additive best-first variants, weighted A* (Hart et al., 1968; Pohl, 1970) and greedy best-first search (Bonet & Geffner, 2001), must follow $g + 3h_{\text{add}}$ and h_{add} order respectively under a declared tie-break. Compliance establishes neither termination, completeness, optimality, nor success.

Trusted Search Runtime. The runtime only validates, applies, persists, or rejects operations and never chooses omitted operands, reorders candidates, or repairs, so raw emissions replay to the same trace (Operation Schemas appendix, Appendix A, and Algorithm Invariant Definitions appendix).

The runtime supports two decision contracts, which differ in what the policy chooses. Under the enumeration contract, each expansion submits its complete candidate set, and the runtime’s priority heap orders it. Under the choice-frontier contract, the policy selects which runtime-owned frontier state to expand next, under a budget of twice the exact reference’s expansion count. The choice-frontier contract runs the two additive best-first variants. Its observation is scene-only, with rendered scenes of the initial state and each frontier state plus goal pages, and opaque labels are the only candidate text, so no observation modality is compared under this contract (Choice-Frontier Contract appendix). Figure 1 and Table 26 (Decision Contracts appendix) compare the two contracts.

Process supervision is token-level cross-entropy cloning of the exact reference’s operation on Search-Trace Segments, bounded slices of replay-verified exact traces. Optional DAGger (dataset aggregation) (Ross et al., 2011) adds the exact reference’s corrections on visited contexts.

4 EXPERIMENTAL DESIGN

Every learned result is read against two oracle-assisted controls, which read the runtime’s valid-operation list or the reference trace, so their scores are bounds, not abilities. The exact reference runs the declared algorithm and defines the imitation target. Where a control saturates, the gain is unmeasurable rather than zero. Under the enumeration contract, random-valid is uniform over valid operations, drawing one listed candidate per decision with a seeded generator and no retry or repair. Candidates are successor rows under additive search, non-duplicate evaluated successors at an admissible frontier position under BFWS, and unvisited successors under BFS, else a frontier-

Box 1: Validate before score. A four-step procedure that admits random-valid as a choice control for a search-execution interface and validates the success measure before any policy is scored. Each step names the Results subsection that applies it.

Inputs. The interface as executed, with its candidate set per decision and its ordering and tie-break rule. The exact reference. Random-valid, seeded, with no retry or repair.

1. Identity audit. Run random-valid and the exact reference on the same tasks and seeds and compare them decision by decision. Report the divergent fraction and the first divergence. Success reads as choice only where decisions diverge and random-valid’s success falls below the reference’s. Otherwise it measures validity, which saturation alone cannot reveal (Section 5.1).

2. Budget check. Apply the compute-matched-baseline principle (Li & Talwalkar, 2020) when a zero-token control exists. Count the budget in decisions or expansions, the units a zero-token control spends, and report control success at the reference’s count and at twice it. The check fails if the control saturates in its native unit but not when counted in decisions (Section 5.4).

3. Ladder. Given a reference that computes a priority order, and on a fresh panel frozen before evaluation, pre-register the order exact reference, ϵ 0.25, 0.50, 0.75, random-valid, where each exact- ϵ rung replaces a fraction ϵ of reference decisions with random-valid choices. M1 is the area under the solve-versus-budget curve up to twice the reference’s expansion count. The ladder passes only if all four adjacent paired task-clustered 95% lower bounds are positive along the ϵ axis. The rungs are privileged, so a pass validates the measurement, not a policy (Section 5.2).

4. Report. Give the policy’s difference from random-valid with a paired task-clustered interval and a pre-registered margin (margin 0.05 here), its ladder position, per-task signs, screened and unscreened panels, and training seeds. Non-separation from a rung is not equivalence (Section 5.3).

Where a step was at risk (one audited instance each). Complete-set submission with a priority heap and sorted tie-breaking failed step 1 (our enumeration contract’s additive cells, 48/48). Per-step choice among valid actions passed step 1 (ScienceWorld). Model-scored candidates on a frontier that keeps every alternative, under a token budget, failed step 2 by construction (LLM-First Search).

If a step fails, change the contract (for example, let the policy choose the frontier state, or break ties in submission order) and audit again.

retire operation. Under the choice-frontier contract, random-valid is uniform over the offered frontier states.

The identity audit compares random-valid with the exact reference decision by decision. It is the choice-frontier contract’s pre-registered pass condition over 18 seed-17 pairs, passing on any divergence without re-runs, and a diagnostic on the enumeration contract’s additive cells.

The validation and held-out panels are frozen before any evaluation and pass a random-valid screen, which the unscreened panel skips. The first measurement used the reused 9-task panel, low-cost tasks from 8 domains shared with the corruption suite, native arms, menu manipulation, and comparator zoo, so each design step could adapt to earlier results. The ladder test and the adapter use the validation panel, 12 fresh tasks from 7 domains. The screen gives each candidate task 10 random-valid episodes and admits it only if 1 to 9 reach the goal, so admitted tasks are ones on which near-uniform choice sometimes succeeds.

The held-out panel (P2 in figures and tables) and the unscreened panel (P2u) were frozen from new seeds before any evaluation episode. Held-out means this panel only, built with the validation panel’s rules from seeds on which no adapter had been evaluated. Its seed extension followed a controls-only screen of the first 40 seeds. The unscreened panel skips the screen and draws kept candidates outside the held-out panel. The frozen 45-task final evaluation remains unexecuted.

Each choice-frontier result has a rule frozen before its outcome. The ladder test passes only if all four adjacent paired 95% lower bounds are positive along the exact reference, the exact- ϵ rungs, and random-valid. Its intervals are task-clustered, resampling whole tasks (seed 133, 10,000 draws). The adapter rule requires adapter minus random-valid M1 to be positive with a materiality margin of 0.05 and sets an equivalence margin of ± 0.05 . Later frozen protocols cover three seeds, two fresh panels, a zero-shot policy, and a second external audit.

The enumeration-contract study compares process SFT with the pretrained base, SFT plus two DAGger iterations (Ross et al., 2011), and an exposure-matched continued-SFT control in each cell. SFT trains a fresh low-rank adaptation (LoRA) adapter (Hu et al., 2022) for one epoch on 512 replay-verified records, and the BFWS gate checkpoint used two epochs on 47,780 records. The backbone

is Qwen3-VL-8B-Instruct (Bai et al., 2025) (revision 0c351dd0), with InternVL3.5-8B (Wang et al., 2025) second. All runs share a 336 GPU-hour cap.

The expanded baseline is the enumeration-contract evaluation grid of text, visual, and multimodal observations over 24 tasks from 12 domains, giving 288 episodes per arm. Each of its cells is one algorithm under one observation. Budgets count decision calls, charge invalid operations, cap adapters at twice the exact reference’s count, and price the base at one call. An episode ends at its first invalid emission and counts only after independent replay. Contrasts use a paired whole-instance bootstrap (10,000 resamples, seed 1729, Statistics appendix).

Every executed run reconciles with its ledger and independent replay (Gate Receipts appendix, Appendix J, Table 3). The frozen 45-task final evaluation was never accessed. Every enumeration-contract learned cell is single-seed except two replication cells (seeds 17/29/71).

Box 1 collects these rules into one procedure, validate before score, and each Results subsection reports one of its steps.

5 RESULTS

Every result below is development-stage, each measurement is kept separate, and the controls are oracle-assisted bounds rather than abilities (Section 4). Section 5.1 applies step 1, Section 5.2 step 3, Section 5.3 step 4, and Section 5.4 steps 1 and 2 on published interfaces. The choice-frontier budget binds by design (Section 4).

5.1 THE IDENTITY AUDIT AND WHAT IT CHANGES

On the enumeration contract’s additive cells, random-valid never decides differently from the exact reference. Compared decision by decision on every additive cell of the expanded baseline, the two were decision-identical on 48/48 pairs, so success there measures operation validity rather than choice. The identity follows from the executed serial rule, under which equal-priority ties break in sorted candidate order, which the Counterfactual appendix confirms by re-running the audit under both orders (Appendix S). The contract specification does not reveal this rule, and the per-decision audit or a reading of the executed controller does. The identity belongs to this controller, not to additive best-first search in general (Algorithm Invariant Definitions appendix). The BFS and BFWS cells register choice, meaning that random-valid’s decisions differ from the reference’s and its success falls below it (Primary-Matrix appendices, Appendices K and L).

Changing only the tie-break order reverses the verdict. When equal-priority ties break in submission order, 19/48 pairs diverge, and the pre-registered prediction of divergence on most pairs (more than 24/48) failed. On the other 29 pairs every equal-priority tie breaks the same way under both orders. Under the choice-frontier contract the audit diverges on 18/18 pairs (Figure 1). Divergence location and per-pair details are in the Counterfactual appendix (Appendix S).

On the expanded baseline, process SFT succeeds on 125/288 entries against 240/288 for random-valid, all in the additive cells that the audit reads as validity, as is the seed-replication gain over the base (Statistics appendix, Table 5). Process SFT scores 0/24 in every BFS and BFWS cell and wins none of the 72 problem-modality units (one task under one observation) in these descriptive single-seed bounds (Primary-Matrix appendices, Appendices K and L). The one positive learned contrast is a fixed-seed run on the frozen text-state BFWS development panel, at 0.933 against 0.467 for random-valid with a paired bootstrap 95% lower bound of 0.2 (Table 6). Saturation alone does not separate the cases, since random-valid saturated the BFS development panel while its expanded-baseline BFS decisions diverge. The Results Detail appendices report failure calibration, corruption, menu manipulation, image-only arms, and secondary panels outside the choice-quality claim. Before random-valid success is read as choice, step 1 of Box 1 must therefore run on each interface’s executed code.

5.2 CHOICE-FRONTIER CONTRACT AND INSTRUMENT VALIDATION

The choice-frontier contract passes the identity audit on the reused 9-task panel, where random-valid diverged from the exact reference on 9/9 tasks (Comparator-Zoo Receipts appendix, Appendix R).

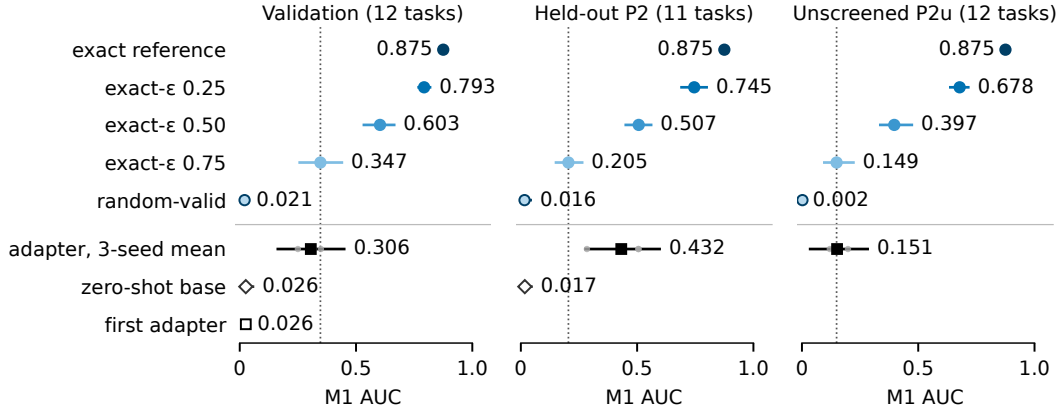


Figure 2: M1, the area under the solve-versus-budget curve, with task-clustered 95% intervals on the validation panel, the held-out panel (P2), and the unscreened panel (P2u). Rows are the five ladder arms, the three-seed adapter mean with faint per-seed points, the zero-shot base (validation and held-out panels), and the first adapter (validation panel only). The rungs fall in order on every panel, the dotted line marks each panel’s 0.75 rung, and only on the held-out panel does the adapter’s interval lie entirely right of it. The rungs read privileged information, so the ladder validates the measurement rather than a policy, and the unscreened ladder is descriptive.

On a fresh validation panel of 12 tasks from 7 domains, the pre-registered ladder test passes. M1 orders the exact reference (0.875) above the three exact- ϵ rungs in order, with random-valid (0.021) lowest (Figures 1 and 2). The reference’s 0.875 is fixed by construction of M1, since goal selection is one decision past the reference’s expansion count (Choice-Frontier Contract and Comparator Selectors appendix). Every adjacent paired task-clustered 95% lower bound is positive, and the smallest gap is $+0.082 [+0.051, +0.111]$. The Choice-Frontier Validation appendix (Appendix T, Table 18) gives panel construction, every arm’s receipts, and all four paired gaps.

The ladder test also passes on the held-out panel and, descriptively, on the unscreened panel. The held-out panel has 11 tasks from 8 domains, built with the validation panel’s rules on fresh seeds. After the first freeze admitted 6 of the 8 required tasks, an amendment made before any evaluation episode added seeds with the rules unchanged (Held-Out Panels appendix, Appendix U.1).

The smallest held-out adjacent gap is $+0.130 [+0.070, +0.190]$. A pass licenses reading a higher M1 as choice closer to the reference’s along the ladder’s ϵ axis (Box 1, step 3). The ladder does not validate M1 off this axis, and a privileged selector that reaches the goal in fewer decisions scores above the reference (Choice-Frontier Contract and Comparator Selectors appendix, Appendix I). The rungs read the reference’s computed priority order, which is never shown to a policy, so the ladder validates the measurement, not a policy. The choice-frontier budget is twice the reference’s expansion count, and the arms table reports random-valid success at that cap (Table 18).

5.3 POLICIES SCORED BY THE MEASUREMENT

The admitted measurement scores one trained policy above random-valid and two policies at its level (Box 1, step 4). Across three training seeds, the trained adapter’s D_3 is positive on the validation panel, meeting the pre-registered rule with every seed positive, and on the held-out panel (Table 1, Scaled Adapter appendix, Appendix U). Seeds 29 and 71 were trained later with the byte-identical recipe under a pre-registered protocol.

The adapter’s ladder position depends on the panel. It is separated above the 0.75 rung only on the held-out panel, $+0.227 [+0.053, +0.403]$. It is not separated from the rung on the validation panel, the unscreened panel, or pooled over 35 tasks, and with no pre-registered equivalence margin, non-separation is not equivalence. On the validation panel it also falls below the 0.50 rung (descriptive). The held-out separation rests on 11 task clusters, where percentile intervals can under-cover, and per-task differences there are positive on 8 tasks and negative on 3.

Two policies score at the random-valid level. Prompted zero-shot under the same contract, observation, and cap, the base model is equivalent to random-valid on the validation panel (development

Table 1: Adapter position on the three choice-frontier panels and pooled over their 35 tasks. Each cell gives the three-seed M1 difference, its paired task-clustered 95% interval, and its verdict. Rows marked descriptive carry no pre-registered rule. Non-separation is not equivalence, since no equivalence margin was pre-registered. The held-out separation rests on 11 task clusters, where percentile intervals can under-cover. Pooled D_3 is in Table 21.

Panel	Adapter minus random-valid (D_3)	Adapter minus exact- ϵ 0.75 rung
Validation	+0.285 [+0.140, +0.433], positive	-0.041 [-0.187, +0.102], not separated
Held-out	+0.416 [+0.250, +0.591], positive	+0.227 [+0.053, +0.403], separated above
Unscreened (descriptive)	+0.149 [+0.030, +0.283], positive	+0.002 [-0.145, +0.157], not separated
Pooled (descriptive)	-	+0.058 [-0.040, +0.156], not separated

stage), with $D_3 = +0.005 [-0.025, +0.044]$ for this single realisation within the pre-registered ± 0.05 margin. On the held-out panel the verdict is inconclusive (confirmatory). It falls below the 0.75 rung and below the trained adapter on both panels (Zero-Shot Policy appendix, Appendix U.3). The first adapter, an earlier small-corpus adapter that differs in corpus, augmentation, sample order, and seed, reaches M1 0.026 [0.010, 0.042] on the validation panel, at the random-valid level.

Without the screen, the gain survives but rests on 4 of 12 tasks. The unscreened panel is drawn from kept candidates outside the held-out panel, and all 12 of its tasks had 0/10 random-valid screening goals. Both panels were frozen from new seeds before any evaluation episode (Section 4). The ladder passes there (descriptive), and the adapter still beats random-valid (Table 1), with the other 8 tasks unsolved by both the adapter and random-valid. This is descriptive evidence that the gain survives without the screen, not evidence about where choice discriminates.

Over all three panels most per-task differences are positive (22 of 35 tasks, with 12 zero and the remaining task negative, descriptive). Leaving out one domain at a time over the validation and held-out panels keeps every D_3 positive with every lower bound positive (descriptive, Held-Out Panels appendix, Appendix U.1). One on-policy DAGger round, run for one seed on half the planned corpus, does not separate the adapter from the 0.75 rung, +0.115 [-0.026, +0.258] over the validation and held-out tasks (DAGger Ablation appendix, Appendix U.2). The choice-quality claim is development-stage, holds relative to random-valid and the exact- ϵ ladder on three panels with three training seeds, and carries no planning-ability or generality claim.

5.4 STEPS 1 AND 2 ON PUBLISHED INTERFACES

ScienceWorld is a positive control for step 1, and LLM-First Search is a by-construction illustration of step 2 (Box 1). On ScienceWorld (Wang et al., 2022), a published sequential-choice interface, five random-valid seeds ran on 200 tasks under the paper’s 100-step budget. Random-valid diverges from the gold trajectory at the first decision in 1000/1000 pairs. Success is 0.915 for the reference versus 0 for random-valid, so success there measures choice. The audit covers 6 of its 30 task types. None of the seven audited-candidate interfaces we surveyed uses an enumeration contract, so the 48/48 identity failure is shown only on our runtime (Table 24 in the External Identity Audit appendix, Appendix V).

On LLM-First Search (Herr et al., 2025), the model scores candidates that a search then orders, and the audit covers 400 pairs over 80 tasks from four configurations. The reference model is a local open-weight substitute (Qwen3-30B-A3B-Instruct) for the paper’s GPT-4o, whose published success on these configurations is 0.768 (LLM-First Search Audit appendix, Appendix V.1). Random-valid diverges from the reference by the third decision in every pair, yet under the token caps it solves all tasks (1.0) by construction, against 0.825 for the reference. A token cap never binds a chooser that spends no tokens, and because the frontier keeps every alternative, such a chooser is a complete randomized search, here using up to 102,606 expansions against the reference’s 1,300. Counted in expansions, random-valid solves 0.0775 at the reference’s own count and 0.19 at twice it, so the budget must count decisions, as step 2 requires.

6 DISCUSSION AND LIMITATIONS

A search-choice measurement must be validated before it scores policies, and ours is. The identity audit returns both verdicts on one runtime, 48/48 identical pairs under sorted ties and 19/48 divergent pairs under submission order (Section 5.1), and it admits the choice-frontier contract, whose decisions diverge. Its budget counts decisions, which applies the compute-matched-baseline prin-

principle (Li & Talwalkar, 2020). Its M1 then passes the pre-registered ladder on the validation panel and again on the held-out panel (Section 5.2). A higher M1 therefore reads as choice closer to the reference’s along the ladder’s ϵ axis. The rungs read the reference’s computed priority order, which is never shown to a policy, so the ladder validates the measurement, not a policy.

The development-stage answer to the title is that a trained vision-language model (VLM) adapter learns to choose which frontier state to expand in the two additive best-first variants better than random-valid, without approaching the reference. Under the enumeration contract no BFS or BFWS cell shows learned success. The trained adapter’s M1 exceeds random-valid’s on every panel, descriptively on the unscreened panel (Section 5.3). Its M1 of 0.151 to 0.432 stays well below the reference’s 0.875, which is fixed by construction. The adapter is trained only by imitating the exact reference.

Box 1 states three practices for benchmark designers and policy trainers. First, designers should audit decisions on the executed runtime and report the divergent fraction. Second, designers should count budgets in decisions or expansions whenever a zero-token control exists (Box 1, step 2). Third, trainers should report the gain over random-valid with its ladder position, split by audit verdict. A gain over the base is not choice, as the seed-replication cell shows (Section 1). A gain over random-valid is not near-reference choice unless it separates from a rung.

ScienceWorld (Wang et al., 2022) is a positive control, and LLM-First Search is a by-construction illustration of the budget step (Section 5.4). None of the seven audited-candidate interfaces we surveyed uses an enumeration contract, so the 48/48 identity failure is shown only on our runtime.

Limitations.

- *Ladder position.* The trained adapter is separated above the 0.75 rung only on the held-out panel, not on the validation panel, the unscreened panel, or pooled.
- *Ladder axis.* The ladder validates M1 only for mixtures of the reference with random choice, and a privileged selector that reaches the goal in fewer decisions scores above the reference.
- *Screening.* The held-out panel is screened like the validation panel, and on the unscreened panel the gain rests on 4 of 12 tasks.
- *Seeds.* There are three training seeds, and seed 29 is weakest on every panel (descriptive).
- *Final evaluation.* The frozen 45-task final evaluation is unexecuted.
- *Prediction.* The pre-registered majority prediction for submission-order divergence failed.
- *Runtime.* The runtime never chooses or repairs operands, so no result shows the model searching without the runtime’s validation.
- *External audit.* The LLM-First Search reference uses a substituted local model, and the audit covers 80 tasks from four configurations.
- *Second policy.* The second policy scored is zero-shot, and the trained second backbone was not run under the choice-frontier contract (it ran at reduced scope under the enumeration contract). The measurement’s ability to order two trained choosers that both beat random-valid is therefore untested.
- *Menu surface.* Our manipulations of the enumeration contract’s action menu cannot separate surface-following from state grounding, which only the unrun test comparing fixed with random distractor positions would do.
- *Modality.* The choice-frontier results use one scene-only observation, so no modality is compared under that contract.

7 CONCLUSION

We validate a search-choice measurement before it scores any policy. An identity audit and a decision-counted budget admit the interface, and a pre-registered privileged ladder validates M1 on two panels. Scored this way, a trained adapter beats random-valid for frontier selection in the two additive best-first variants ($D_3 = +0.285 [+0.140, +0.433]$ on the validation panel), without approaching the reference. Separation above the 0.75 rung holds only on the held-out panel. The choice-quality claim is development-stage, holds relative to random-valid and the exact- ϵ ladder on three panels with three training seeds, and carries no planning-ability or generality claim.

REPRODUCIBILITY STATEMENT

Box 1 states the validate-before-score procedure, and each Results subsection reports one of its steps. Reproduction details are in the Trusted Search Runtime (Appendix C), Data and Split Procedures (Appendix D), Receipt Tables (Appendix E), and Statistics (Appendix F) appendices. The Choice-Frontier Contract and Comparator Selectors appendix (Appendix I) specifies the choice-frontier contract.

The Counterfactual (Appendix S), Choice-Frontier Validation (Appendix T), Scaled Adapter (Appendix U), and External Identity Audit (Appendix V) appendices report the follow-ups. The Held-Out Panels (Appendix U.1), DAGger Ablation (Appendix U.2), and Zero-Shot Policy (Appendix U.3) appendices sit under the Scaled Adapter appendix, and the LLM-First Search Audit appendix (Appendix V.1) sits under the External Identity Audit appendix. Table 3 holds the ledgers, and Table 4 indexes the pinned evidence paths. Code and evidence bundles will be provided as anonymous supplementary material.

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Appendix roadmap. The Operation Schemas (Appendix A), Algorithm Invariant Definitions (Appendix B), and Trusted Search Runtime (Appendix C) appendices show that the runtime never chooses or repairs on the policy’s behalf. The Data and Split Procedures appendix (Appendix D) shows that no evaluation task leaked into training, and the Receipt Tables appendix (Appendix E) traces each reported run to its receipt. The Statistics appendix (Appendix F) states what an interval licenses, and the Choice-Frontier Contract and Comparator Selectors appendix (Appendix I) specifies the measurement that Box 1 validates. The Counterfactual appendix (Appendix S) holds the submission-order audit behind the enumeration-contract identity failure, and the Choice-Frontier Validation appendix (Appendix T) holds the ladder. The Scaled Adapter appendix (Appendix U) and its Held-Out Panels, DAgger Ablation, and Zero-Shot Policy subsections (Appendices U.1, U.2, and U.3) hold the adapter and the policies scored beside it. The External Identity Audit appendix (Appendix V) and its LLM-First Search Audit subsection (Appendix V.1) hold the interface survey and both external audits. The Primary-Matrix appendices (Appendices K and L) and the other Results Detail appendices show what the enumeration-contract learned results measure. The Decision Contracts appendix (Appendix X) compares the two contracts.

A OPERATION SCHEMAS

This appendix gives the operation vocabulary, validity checks, and rejection kinds of each declared algorithm, for readers who need the exact interface behind the Typed Search Operation definition.

Each declared algorithm defines a vocabulary of operation types τ . A policy emission $u_t = (\tau_t, \theta_t)$ pairs a type from that vocabulary with an operand vector θ_t carrying every exploration-determining argument. The runtime rejects any emission whose type falls outside the declared vocabulary or whose operand vector omits a required field. Every operand that determines exploration must originate from the policy (Section 3).

BFS. The primary operation is node expansion. Its operands name the source state (the current frontier head), the grounded successors selected for insertion, the action producing each successor, and the enqueue position for each. The runtime checks that the named source matches the FIFO frontier head, that each successor results from an applicable grounded action in the source state, that the successor is unvisited, and that each insertion targets the FIFO tail. A goal-test operation names the state to test. The BFS pilot corpus exposes exact grounded successor candidates and full visited membership at each step.

BFWS (complete unpruned). Expansion operands carry the novelty value and goal-count priority of the source state alongside the fields above. Insertion operands carry each candidate’s duplicate membership, novelty, and priority. The runtime checks that the selected source has the declared best priority, that duplicate membership is correctly reported, and that insertions follow the declared ordering. The BFWS corpus exposes candidate duplicate, novelty, priority, insertion, and enqueue facts at every training observation.

Additive best-first. Operands carry the source state’s accumulated path cost g , heuristic estimate h_{add} , and composite value $f = g + w \cdot h_{\text{add}}$ with $w = 3$ for the weighted variant, or $f = h_{\text{add}}$ for the greedy variant. Insertion operands carry per-successor g , h_{add} , and f . The runtime checks that the expanded state has the minimum f in the frontier under the declared tie-breaking rule and that all arithmetic is consistent.

Validity preconditions. An emission passes through a fixed sequence of checks. Failure at any step rejects the emission with the corresponding kind and terminates the episode. The checks run in the following order.

1. **Parse.** The raw output must parse into a well-formed Typed Search Operation with a recognized type and complete operand vector. A failure is recorded as `malformed_output`.
2. **Source identity.** The named source state must match the state the algorithm expects. A failure is recorded as `source_state_mismatch`.
3. **Action applicability.** Every grounded action in the operands must be applicable in the source state. A failure is recorded as `inapplicable_grounded_action`.

Table 2: Rejection taxonomy from the failure-mechanism calibration. Counts are failures per arm class across 2,910 replay-verified episodes (905 total failures, 0 unclassified). Share is within the arm class.

Arm class	Rejection kind	Failures	Share
Base (447 ep., 447 fail.)	malformed.output	379	84.8%
	invalid.frontier.operation	67	15.0%
	other.invariant.violation	1	0.2%
Learned (906 ep., 407 fail.)	other.invariant.violation	229	56.3%
	malformed.output	72	17.7%
	inapplicable.grounded.action	56	13.8%
	effect	27	6.6%
	schema	16	3.9%
	invalid.frontier.operation	5	1.2%
	source.state.mismatch	1	0.2%
	static.context	1	0.2%
Oracle control (51 fail.)	expansion.budget.exhausted	51	100.0%

4. **Successor schema.** Any named successor state must satisfy the domain schema. A failure is recorded as `schema`.
5. **Successor effects.** The named successor must match the declared effects of the applied action. A failure is recorded as `effect`.
6. **Static context.** Referenced predicates, objects, and constants must be consistent with the task instance. A failure is recorded as `static.context`.
7. **Frontier discipline.** The frontier manipulation must respect the declared ordering (FIFO for BFS, priority for BFWS, *f*-order for additive best-first). A failure is recorded as `invalid.frontier.operation`.
8. **Algorithm invariant.** Any remaining algorithm-specific constraint not covered above. A failure is recorded as `other.invariant.violation`.

The two broad categories are *malformed-output* rejections, where the emission never reaches the operation checker, and *post-parse rejections*, where the emission parses but fails a runtime check. The oracle-assisted random-valid control, which emits only runtime-accepted operations, has a single failure kind: `expansion.budget.exhausted`, recorded when the frozen episode budget runs out before a goal is reached.

Rejection counts. Table 2 gives the counts from the failure-mechanism calibration, which classified all 905 failures across 2,910 replay-verified episodes with zero unclassified. The `effect`, `schema`, and `static.context` kinds arise in successor-prediction episodes, where the model generates a dynamic-state successor.

B ALGORITHM INVARIANT DEFINITIONS

This appendix states the invariant the runtime enforces for each declared algorithm, which Section 3 defines only in outline.

The Trusted Search Runtime checks the following invariants on every operation. Stepwise validity and invariant compliance do not establish termination, completeness, optimality, or episode success (Section 3).

BFS. The frontier is a FIFO queue. Every expansion must select the queue head as its source. Every insertion must target the queue tail. A state already in the visited set must not be re-inserted. This invariant is sufficient for completeness in finite state spaces with unit-cost transitions.

BFWS (complete unpruned). The frontier is a priority queue ordered first by novelty rank (lower is better), then by goal count among states with equal novelty. No generated successor is pruned. A successor whose state already appears in the frontier or visited set is not re-inserted.

Additive best-first, weighted ($w = 3$). The frontier is a priority queue ordered by $f(s) = g(s) + 3 \cdot h_{\text{add}}(s)$, where $g(s)$ is accumulated path cost and $h_{\text{add}}(s)$ is the additive heuristic. Among

equal- f states the lower generation serial is expanded first. The expanded state is the heap head. A successor with lower g than an existing frontier entry for the same state reopens that entry with improved cost.

Additive best-first, greedy. The frontier is ordered by $h_{\text{add}}(s)$ alone, with no path-cost component, then by ascending generation serial among ties. The expanded state is the heap head. Reopen and duplicate rules follow the weighted variant. In both additive settings, `BestFirstController.start_expansion` assigns serials before submissions in the deterministic `PDDLStateAuthority.applicable_actions_order` (sorted by `GroundedAction.serialize`). The `BestFirstController.apply_operation` heap orders entries by (priority, serial, state identifier, path cost), and `finish_expansion` requires the full candidate set. Thus submission permutations leave frontier order unchanged on the enumeration contract. A runtime assigning serials in submission order would not be submission-order invariant and would not show this identity.

C TRUSTED SEARCH RUNTIME

This appendix gives the runtime’s guarantees and its step and episode pseudocode, for readers checking that the runtime never chooses or repairs on the policy’s behalf.

The runtime provides three guarantees that hold across all declared algorithms.

1. **No-repair property.** The runtime validates, applies, persists to Search Memory, or rejects each emission. It never chooses omitted operands, reorders candidates, silently repairs an invalid emission, or substitutes a default search decision.
2. **Reject-and-terminate.** On the first rejected emission the episode terminates with the rejection kind recorded. There is no retry path.
3. **Replay determinism.** Replaying the raw emissions $u_{1:T}$ against the same task instance and the same initial Search Memory reproduces the explored trace. Any step whose outcome depends on runtime repair would violate this property.

The procedures below specify the per-step check and the episode loop. No `algorithm` package is loaded. The listing uses $\LaTeX$ tabbing.

```

procedure STEP( $u_t, M_{t-1}, A, x$ )
  // Phase 1: Parse the raw emission
  if  $u_t$  does not parse as  $(\tau, \theta)$  with  $\tau \in \text{vocab}(A)$  then
    return REJECT(malformed_output)
  // Phase 2: Validate source identity
  if  $\theta.\text{source} \neq \text{expected}(A, M_{t-1})$  then
    return REJECT(source_state_mismatch)
  // Phase 3: Validate operands against the task and state
  for each grounded action  $a$  named in  $\theta$  do
    if  $a$  is not applicable in  $\theta.\text{source}$  under  $x.D$  then
      return REJECT(inapplicable_grounded_action)
  for each successor state  $s'$  named in  $\theta$  do
    if  $s'$  violates the domain schema then
      return REJECT(schema)
    if  $s' \neq \text{effects}(a, \theta.\text{source})$  then
      return REJECT(effect)
  if any static-context reference in  $\theta$  is inconsistent with  $x$  then
    return REJECT(static_context)
  // Phase 4: Check Algorithm Invariant
  if frontier discipline of  $A$  is violated by  $(\tau, \theta, M_{t-1})$  then
    return REJECT(invalid_frontier_operation)
  if any other invariant of  $A$  is violated then
    return REJECT(other_invariant_violation)
  // Phase 5: Apply and persist
   $M_t \leftarrow \text{apply}(\tau, \theta, M_{t-1})$ 
  return ACCEPT( $M_t$ )

```

```

864 procedure EPISODE( $x, A, \phi_m, B$ )
865    $M_0 \leftarrow \text{InitSearchMemory}(A, x)$ 
866   for  $t = 1, 2, \dots, B$  do
867      $o_t \leftarrow \phi_m(c_t(x, M_{t-1}))$  // build observation
868      $u_t \leftarrow \pi(o_t)$  // policy emission
869      $r \leftarrow \text{STEP}(u_t, M_{t-1}, A, x)$ 
870     if  $r$  is REJECT( $kind$ ) then
871       return ( $t, \text{FAIL}, kind$ )
872      $M_t \leftarrow r.M$ 
873     if GoalTest( $M_t, x.G$ ) = TRUE then
874       return ( $t, \text{SUCCESS}$ )
875   return ( $B, \text{FAIL}, \text{expansion\_budget\_exhausted}$ )

```

Each call to STEP consumes one unit of the frozen episode budget B , whether the emission is accepted or rejected. Invalid operations are therefore charged to the budget (Section 4).

D DATA AND SPLIT PROCEDURES

This appendix describes the corpora, splits, and evaluation manifests behind the reported panels, for readers checking that no evaluation task leaked into training.

All training and evaluation splits are by whole task instance. No state, decision, or target from an evaluation task appears in any training row. Every released corpus passed no-overlap, no-conflict, live/training-parity, and token-budget audits before use.

BFS pilot corpus. The v6 text-state process corpus holds 25,109 process rows split into 12,994 train and 12,115 dev rows, drawn from 90 replay-verified trusted traces covering 90 distinct tasks. The release assigns splits by semantic task identity, with zero cross-split task overlap. The v8 evaluation panel selects 15 tasks from this corpus’s development set, not its training set. No v8 evaluation-panel task contributes training rows, consistent with the whole-instance rule above. Corpus observations expose the canonical task context, exact grounded successor candidates, and full visited membership. Release audits recorded zero semantic, input, and input-target overlap, zero conflicting identical-input labels, zero teacher-operation rejections, and zero live/corpus mismatches, all within an 8,192-token input cap and a 384-token output allowance.

BFWS corpus. The corpus covers the frozen 105-task development panel with 69,019 replay-verified teacher decisions. Its release contains 69,019 process records, 67,215 operational records, and a process projection of 47,780 train and 21,239 dev rows. Each step’s observation exposes candidate duplicate membership, novelty values, goal-count priorities, insertion positions, and enqueue facts. Audit counters recorded zero overlap, mismatch, rejection, or final-evaluation-access violations.

Evaluation manifests. The BFS pilot uses a frozen 15-task outcome-blind development panel. The BFWS structural gate froze a 105-task development panel and, separately, the manifest of the frozen 45-task final evaluation. That manifest has never been accessed, and no model prediction, training decision, or analysis references it. The expanded baseline evaluates 24 tasks from 12 domains under 4 algorithms, 3 modalities, and 4 arms (1,152 logical bindings, 576 model episodes, all independently replayed). The curriculum stage evaluates 27 unseen tasks. The generalization-v2 protocol admits 25 derived tasks from 93 eligible semantics-preserving variants (uniform $k=5$ cheapest-prefix per family, membership frozen before execution).

E RECEIPT TABLES

This appendix holds the gate receipts, the per-window ledgers, and the artifact index, so that a reader can trace each reported run to its receipt.

Table 3 combines branch-level frozen scope, gate status, coverage, and compute from the synthesis and follow-up ledgers. Five branches completed at v1 scope. Two stopped at frozen v1 cost-

Table 3: Combined per-branch gate receipts and ledger allocations. The expanded program spent 68.52 of 336 GPU-h, rounded independently by row. The second-backbone cap reflects a prospective 12.11 GPU-h transfer from recovery reserve. Corruption and seed replication draw on existing branch caps. Follow-up windows have separate ledgers and are never summed into the program total or with each other. The R1-R4 spend covers both stress-test rows.

Branch	Tickets	Terminal state	Frozen scope / replay coverage	GPU-h / cap
Expanded baseline	#116–#118	complete	1,152 bindings, 576 model, 1,152/1,152 replayed	6.53 / 56
DAGger	#78–#84	complete	405 episodes + 81 paired rows, 405/405	10.91 / 48
Successor prediction	#85–#89, #99	complete	90 episodes, 15 cells, 90/90	10.85 / 64
Curriculum by modality	#119	complete	243 model + 324 comparator bindings, 567/567	11.78 / 48
Transfer	#104–#107	complete	39 cells (12 v1, 27 v2), 39/39	3.60 / 24
Second backbone	#101–#103, #123	v1 VALID_STOP, v2 complete	72 model + 72 comparators, 144/144	6.03 / 52.11
Generalization / robustness	#96, #98, #124	v1 VALID_STOP, v2 complete	1,200 episodes (300 model, 900 controls), 1,200/1,200	6.04 / 32
Observation corruption	#126	complete	360/360 new bindings replayed	within generalization
Training-seed replication	#127	complete	534/534 new episodes	12.79 / –
Recovery reserve	–	reduced	–	0.00 / 11.89
Total				68.52 / 336
<i>Follow-up windows (separate ledgers, not summed with the total)</i>				
Native arms	#128, #129	v1 gate history, v2 complete	216/216 new + 72/72 comparators	3.2086 / 12
Stress tests R1/R2/R4	#130	complete	72 menu + 80 variant + 18 cross-cell, 170/170	7.7998 / 36
Second backbone v3 (R3)	#130	complete	72/72 model + 72/72 comparators	same window
Choice frontier	#132	CHOICE_SENSITIVE	144/144 panel, 58 preparation episodes re-derived	1.069 / 12
Comparator zoo	#133	complete	432/432 bindings, 342 new	CPU only
Counterfactual audit, M4 intervals	#134	prediction failed	48 pairs, 288 counts + 96 sequences re-validated	CPU only
Instrument validation	#135	PASS	696/696 CPU + 48/48 model episodes replayed	1.19 / 4
Scaled adapter	#136	POSITIVE	24/24 learned episodes replayed	4.84 / 30
External audit	#137	CHOICE_REGISTERED	1000 pairs over 200 tasks	CPU only
Three-seed replication, held-out panels (v4 window)	#138, #139	POSITIVE	48/48 new learned episodes, 638/638 P2 and 696/696 P2u zoo episodes replayed, and 88/88 P2 plus 96/96 P2u learned episodes	15.22 / 24
DAGger ablation (v5 window)	#140	NOT_SEPARATED	24/24 validation-panel and 22/22 P2 episodes replayed	9.06 / 40
Zero-shot second policy (v6 window)	#141	EQUIVALENT (validation), INCONCLUSIVE (P2)	46/46 model episodes replayed	2.25 / 30
LLM-First Search audit (external v2)	#142	SATURATED (paper budget)	80/80 reference and 400/400 random-valid episodes replayed	2.89 / 12

admission gates with terminal VALID_STOP receipts and later completed at reduced v2 scope. Every branch reconciles with zero missing evidence. The Tickets column indexes internal ticket numbers.

At the synthesis snapshot, the program recorded 135 attempts (96 succeeded, 36 failed, 3 cutoff). Failed and cutoff attempts retain their hours in the ledger. Every attempt ended before the absolute scheduler cutoff of 2026-09-21T11:55:19Z.

The observation-corruption suite added 2.22 GPU-h to the generalization branch. The training-seed replication addendum (closed 2026-09-21) added 12.79 GPU-h and 534 independently replayed episodes covering two cells. Together these addenda brought the program total to 68.52/336 GPU-h. The Statistics appendix reports the seed-variance results.

Follow-up windows. Each follow-up window keeps its own ledger, and none is added to the program’s 68.52/336 GPU-h. The native-arms window (v1 probes, then v2 training and evaluation) spent 3.2086 of 12 GPU-h. Its 216 new bindings (144 model episodes and 72 CPU random-valid episodes) and 72 reused comparators were all independently replayed. The R1-R4 follow-up window spent 7.7998 of 36 GPU-h across two ledgers. It was resized from 20 GPU-h before any outcome, because its frozen worst-case estimate of 27.04 GPU-h including spend exceeded the initial cap, and realized cost landed at 29% of that estimate. The choice-frontier window spent 1.069 of 12 GPU-h (training 0.81, smoke 0.052, model evaluation 0.208), with corpus derivation and controls on CPU. Its 144 panel episodes were replayed and its 58 preparation episodes re-derived bit-exact. The comparator zoo launched no GPU job. It replayed all 432 zoo bindings, of which 342 are new CPU episodes and 90 are reused 2× random-valid controls. The three-seed replication and the held-out panels shared one window, which spent 15.22 of 24 GPU-h, and the DAGger ablation window spent 9.06 of 40 GPU-h, including 1.79 GPU-h charged for a cancelled collection. The zero-shot window

Table 4: Artifact index for the follow-up evidence. Each row names a pinned path and the claim class that rests on it.

Pinned path	Claim class
outputs/native-arms/v1/evaluation/ (evaluation.json, analysis.json)	Native-arms clean and corruption success, paired contrasts
outputs/native-arms/v1/identity-audit. json	Identity finding on the enumeration contract
outputs/native-arms/v1/ stress-evaluation.json	Menu manipulation, degraded strains, R2 cross cell, stress replay
outputs/native-arms/v1/r2-decomposition. json	Text-corruption termination modes
outputs/native-arms/v1/budget.json and the R1-R4 window ledgers (v1, v2)	Native-arms and R1-R4 window ledgers
outputs/expanded-study/v1/ second-backbone-v2/evaluation/ outputs/expanded-study/v1/ second-backbone-v3/evaluation/ outputs/choice-frontier/v1/evaluation/ (evaluation, identity-audit, analysis, final-audit)	Second-backbone BFS cell Second-backbone additive replication and its contrasts Choice-frontier gate, panel rows, contrasts, re- play
outputs/choice-frontier/v1/preparation/ visual-choice-frontier/ (report, audit)	Choice-frontier corpus derivation
outputs/choice-frontier/v1/budget.json	Choice-frontier window ledger
outputs/choice-frontier/o4/zoo/manifest. json	Zoo coverage and replay
outputs/choice-frontier/ o4/metrics/ (analysis, stored-posthoc-analysis, optimal-costs, native-four-algorithm-reference-costs)	Exploratory M1–M5 metrics, Holm family, optimal costs, native reference costs
docs/experiments/native-arms/ issue-129-closeout.md, issue-130-closeout.md	Native-arms and stress-test R1-R4 records
docs/experiments/choice-frontier/ issue-132-closeout.md	Choice-frontier outcome and its binding erra- tum
docs/experiments/choice-frontier/ metric-adjudication-2026-09-23.md	Post-close metric adjudication
outputs/choice-frontier/v4/seeds/ metrics/analysis.json	Three-seed endpoint, separation, seed variance on the validation panel
outputs/choice-frontier/v4/panels/ metrics/analysis.json	Held-out and unscreened panels, per-panel arms, pooled results, concentration
outputs/choice-frontier/v5/metrics/ analysis.json	DAGger ablation
outputs/choice-frontier/v4/budget.json, outputs/choice-frontier/v5/budget.json	v4 and v5 window ledgers
outputs/choice-frontier/v6/metrics/ analysis.json	Zero-shot second policy
outputs/choice-frontier/v6/budget.json	v6 window ledger
outputs/external-audit/v2/lfs/ identity-audit.json	LLM-First Search audit
outputs/external-audit/v2/budget.json	External audit v2 ledger
Comparator-zoo ticket (#133) closing comment	Zoo numbers table and claim boundary

spent 2.25 of 30 GPU-h, and the LLM-First Search audit spent 2.89 of 12 GPU-h on its reference server with no paid API use. Failed attempts retain their hours in every window.

Artifact index. Table 4 maps each pinned follow-up evidence path to the class of claim it supports. Paths are relative to the evidence repository.

F STATISTICS

This appendix states the interval, test, margin, and tiny-strata conventions behind every reported contrast, for readers judging what an interval licenses.

Default contrast procedure. Contrasts are paired by whole task instance. Paired differences use a bootstrap with 10,000 resamples at seed 1729 and 95% percentile intervals. Under one training seed and panels of 24 or fewer instances, these intervals are descriptive bounds. They do not support broad superiority claims.

The BFWS gate used confidence level 0.95 with 10,000 paired whole-instance resamples (bootstrap seed 1729). Its reported lower bound is the 0.025 quantile, i.e., the lower endpoint of a two-sided 95% percentile interval.

Tiny-strata rule. Strata with fewer than 8 episodes are reported as descriptive only. No intervals are computed over such strata. They are listed explicitly and never silently pooled or dropped.

Transfer tests. Transfer comparisons use McNemar exact tests against the per-example base predictions, with Holm correction across all 36 adapter-by-benchmark comparisons. The minimum adjusted p in the executed transfer branch is 0.8156. None survives correction.

Follow-up bootstrap conventions. The follow-up windows fix their own bootstrap seeds, separate from the main program’s seed 1729 default. The native-arms contrasts and the choice-frontier contrasts use a paired whole-problem bootstrap at seed 61813 with 10,000 resamples and 95% percentile intervals. The second-backbone v3 contrasts (stress test R3) use seed 1729 with 10,000 resamples. The comparator-zoo metrics use a task-cluster bootstrap that draws nine tasks with replacement at seed 133 for 10,000 draws. Both additive cells and all random seeds stay inside their task cluster.

Materiality and non-equivalence. A follow-up contrast is material when its 95% interval excludes zero. The follow-up protocols through the comparator zoo pre-registered no equivalence margin, so a non-material contrast there supports neither superiority nor equivalence. The later choice-frontier protocols pre-register an equivalence margin of ± 0.05 for D and for the DAgger change Δ , and none for the separation S , so a non-separated S is not equivalence to the rung. Wherever the text reports a non-material contrast or a comparison that fails Holm correction, an adjacent sentence states that non-significance is not equivalence. The exact sign-flip p value enumerates all 2^n sign vectors of the n per-task differences, is two-sided, tests a zero difference, and is descriptive, so it never supports equivalence.

Tiny strata in the follow-ups. The fewer-than-8 rule also governs discriminative subsets. The choice-frontier learned-minus-random-valid contrast is computed over 18 cells, but it rests entirely on storage, the only panel task where random-valid’s seed frequency lies strictly between 0 and 1 (0.2). A one-task discriminative subset falls under the rule, so that contrast is descriptive. The zoo’s M5 subset, defined from controls only, holds the same single task and is reported descriptively.

Zoo dominance tests. Any positive learned-versus-comparator dominance claim in the comparator zoo uses paired task-cluster bootstrap one-sided p values with Holm step-down correction over five comparators and three metrics, fifteen tests in all. Undefined or uninformative co-solved contrasts prohibit a positive dominance claim and never count as wins. All zoo metrics are exploratory, and those computed on the stored choice-frontier episodes are post-hoc.

Seed-variance addendum. A prospective training-seed replication measured variance for exactly two cells. Seeds 17, 29, and 71 were frozen before any seed-29 or seed-71 artifact existed. All 534 new episodes were independently replayed. Cross-seed aggregation is descriptive ($n = 3$ seeds).

Cell A is the greedy $\times$ multimodal replication cell (`best_first_add_greedy` $\times$ multimodal-state) evaluated on the 24-problem panel. Each seed trained a fresh LoRA adapter with new initialization and dropout randomness on the identical frozen 512-record membership. Table 5 gives the per-seed results.

Cell set B covers DAgger iteration-1 versus the exposure-matched continued-SFT control under BFS across all three modalities. The null replicates, with 0/72 unseen successes for every arm and every seed. The paired contrast (DAgger minus continued SFT) is $+0.000$ $[0.000, 0.000]$ at each seed.

This addendum supersedes the blanket single-training-seed boundary for these two cells only. Every other enumeration-contract cell in the program remains single-seed (seed 17) evidence.

Table 5: Seed-variance replication for Cell A (the greedy $\times$ multimodal replication cell). Contrasts are paired whole-instance bootstrap (24 tasks, 10,000 resamples, seed 1729). The primary contrast (SFT – base) is positive at every seed with no sign flip.

Seed	Successes / 24	SFT – base	SFT – random-valid
17	21/24	+0.875	−0.125
29	21/24	+0.875	−0.125
71	22/24	+0.917	−0.083

G MODALITY ADAPTERS AND OBSERVATION CONTRACT

This appendix defines the observation adapters and what each one exposes to the policy, which bounds any reading of the modality results.

Three fixed Modality Observation adapters are declared. Each maps a decision context c_t (the canonical task context plus the decision-relevant view of Search Memory M_t) to an observation $o_t = \phi_m(c_t)$ presented to the policy. The adapter is fixed for an entire training or evaluation cell. The adapter contract requires decision sufficiency, meaning that o_t must expose every fact the exact reference uses to resolve step t under the declared algorithm.

text-state. The state, goal, and search context are serialized as a compact relational representation. Object identities, predicate names, and all Search Memory contents (frontier, visited set, algorithm-specific state) appear as structured text.

visual-state. The state and goal are rendered as a pair of 128×128 pixel images, a rendered state image and a partial-goal constraint image. Object labels are not embedded in the rendered image. Search Memory contents and the grounded candidate list remain in the textual serialization, even in this adapter. This is the published frozen `visual-state` arm that the native arms below are compared against.

multimodal-state. The observation combines the text-state serialization and the visual-state rendered images. The policy receives the relational text and the image pair jointly.

Native image-only arms. Two further adapters executed under the native-arms v2 contract. `visual-nomem-state` keeps only the static task interface and the grounded, unscored candidate menu as text, and carries state facts in the current rendered scene alone. `visual-seq-state` adds up to $K=8$ rendered history frames, each labelled with its absolute path depth. Neither receives textual Search Memory, accepted deltas, g/h /priority scalars, candidate scores, or membership flags.

Text-scaffold caveat. Search Memory contents and the candidate list are serialized as structured text in all three declared adapters, including visual-state. Rendered images carry the planning state and goal visuals, but search bookkeeping (frontier, visited set, novelty counters, candidate list) always travels through the text channel. The observation-corruption suite is consistent with this. Copyability-preserving text shuffling drops success by up to -0.944 (text) and -0.722 (multimodal), while visual corruption costs at most -0.111 with intervals that include zero.

The native arms narrow this caveat to the candidate menu. With the textual bookkeeping and scalars removed, both arms solve 18/18 clean tasks with zero invalid operations, and blanking or degrading every image of the seq arm gives a corrupted-minus-clean contrast of 0.000 [0.000, 0.000]. As a post-hoc interpretation, the grounded unscored menu is the remaining text channel that could carry the signal. Menu manipulation shows that the policies do not reject schema-valid inapplicable menu entries (pick-rate at chance), and on the additive cells submission-order invariance makes any listed candidate a correct emission. The caveat therefore still holds for every adapter that presents the menu, and no executed adapter isolates an image-only decision channel.

Information-loss boundaries. The 128-pixel unlabelled rendered views can lose symbolic identities. The P3 name-compression family in the generalization-v2 suite is information-lossy. Family-wide (24 eligible P3 variants), 24/24 variants are text-lossy, 18/24 are multimodal-lossy, and 0/24 are visual-lossy. Modality matching between training and evaluation ensures identical observation format, not information parity. Static-context and goal fields of the observation contain text in every

adapter. The available evidence does not establish semantic parity across adapters or an intrinsic modality effect.

H RETAINED-INFRASTRUCTURE FEASIBILITY NOTES

The terms below describe verified infrastructure components that support corpus generation, evaluation, and evidence production. They constitute feasibility evidence for the runtime and rendering pipeline. They cannot support a Search Process Policy efficacy claim.

Plan pipeline. **Plan Submission** transmits a supplied plan from the project adapter to a planning or rendering backend. **Plan Interpretation** converts the submitted plan into the backend’s ordered action representation. An **Action Sequence** is the canonical identity of a supplied plan as a normalized, ordered sequence of grounded action predicates. Casing and whitespace are not part of its identity.

Rendering pipeline. **Render Production** creates a visualisation output and its rendered image for a planning state. **Render Validation** confirms that the produced visualisation and image satisfy declared structural and semantic checks. **Plan Provenance** links a produced visualisation to the exact supplied plan that the backend interpreted.

Certification and evidence. **Integration Certification** is the finding that Plan Submission, Plan Interpretation, Plan Provenance, Render Production, Render Validation, and required execution-isolation checks all passed. An **Attempt** is one authorized, immutable execution with its own identity and retained outcome, whether successful or not. An **Evidence Bundle** is the self-contained records from an Attempt that allow an independent reproducer to verify its claims offline. The retained infrastructure supplies deterministic compact delta/event evidence with semantic replay while preserving search semantics, serving the replay-determinism requirement of the Trusted Search Runtime.

Historical terms. Planning Certificate, Joint Action-and-Certificate SFT, Adaptive Scaffolding, Support Route, Live Memory, Verified Joint Step, and Route Label are historical CGAS terms. They describe retained infrastructure internals. They cannot support any current Search Process Policy efficacy claim and are not referenced elsewhere in this manuscript.

I CHOICE-FRONTIER CONTRACT AND COMPARATOR SELECTORS

This appendix specifies the choice-frontier contract, its corpus derivation, its metrics, and the comparator selectors, for readers reproducing the measurement that Section 5.2 validates.

Menu and observation. At each decision the runtime exposes the live frontier as an unscored menu of opaque labels $c_0, \dots, c_K$. Labels are assigned after a seeded Fisher–Yates permutation of the generation-serial member order, with a per-decision seed derived from master seed 51131 and a hash of the member state identifiers, so a label carries no positional information. The observation holds static task-context pages, the unlabelled 128px initial-state scene, one unlabelled 128px scene per frontier state in menu order, and the partial-goal pages. Its text payload carries only the algorithm name, the label list, the representation tag, the schema version, and the frozen legend. A leak check fails closed on any forbidden key marker.

Output and validation. The policy emits canonical JSON with the single key `expand_choice` and a label value. The runtime rejects malformed JSON, a wrong key set, a non-string label, a label not offered by the pending menu, and a label bound to a non-live member. The first rejection ends the episode as `deterministic_invalid_operation`. Other terminations are `goal_reached`, `frontier_exhausted`, `decision_budget_exhausted`, and `expansion_budget_exhausted`.

Expansion and budget. On acceptance the runtime removes the chosen member from the frontier and expands it atomically. Candidate generation, serial assignment, admission, duplicate detection, reopening, and expansion bookkeeping are inherited unchanged from the frozen additive best-first controller, which is not modified. Per-candidate admissions are runtime-internal, so one submitted choice is one decision. Selecting a member that satisfies the goal solves the task without an expansion and without budget charge, so the exact episode takes its reference expansion count plus one decisions. Decision and expansion caps both equal $2 \times$ the reference expansion count.

Corpus derivation. The training corpus was derived on CPU by replaying the frozen exact traces through the new contract and verified expansion-for-expansion against them (58 episodes bit-exact). The frozen matched-modalities membership indexes enumeration-contract decisions. Each such decision maps to the expansion whose candidate set it submitted into. Unique (task, expansion) pairs in first-occurrence order form the base set, and a round-robin over the frozen task order tops the set up to 512 training records per algorithm. Diagnostics are the unique expansions of the frozen diagnostic identifiers, with no top-up. The imitation target is the label bound to the heap head.

Five fail-closed gates run before training. They check expansion-for-expansion agreement with the stored exact traces, a token budget of measured input plus 384 within 32,768, the text-leak check, image resolution, and identity with the frozen membership. The pinned preparation report records 29 tasks, 1,046 records (512 training and 11 diagnostic per algorithm), 3,544 scenes, and 58 episodes re-derived bit-exact, with outcome PASS. The smoke gate accepted 40/40 model calls.

Training and replay. Each of the two cells trains a LoRA adapter (rank 64, α 128) for 16 updates on its 512 records at seed 17 with the frozen expanded-program recipe. Both training jobs completed all 16 steps and saved final adapters, then failed in post-save summary construction. Their reports were recovered from the primary training-state artifacts with provenance marked, no retraining was needed, and the training audit passes on the recovered reports. At finalize, independent replay re-derives every menu binding, payload, and trusted transition from the stored events, and checks pixel bindings against the retained scenes.

Comparator selectors. Each selector’s choice is mapped back to the menu binding, and every candidate still enters through the additive admission path. Frontier admission, duplicate detection, h_{add} computation, reopening, and serial ordering therefore stay additive for `bfs-order` and novelty-first selection alike. The novelty features of novelty-first selection are positive PDDL state atoms, and its seen-feature sets advance only when a state is expanded, including the initial expansion. Each new episode is re-run against an independently constructed PDDL authority, which checks menus, decisions, admissions, and result dictionaries. Selector episodes replay the model input, menu, actions, admissions, and budget, but they neither generate nor replay image bindings. The optimal costs C^* come from one offline shortest-plan computation per task with replay-verified plans, and the native four-algorithm reference costs are recorded separately from the zoo.

Metric computation. Under M1 an episode solves at multiplier m when its first goal selection occurs within $\lfloor mR_t \rfloor$ decisions with at most $\lfloor mR_t \rfloor$ preceding expansions, where R_t is the additive exact-reference expansion count. Unsolved $2 \times$ runs count as failures at every m , and the same prefix truncation applies to every arm. Task and algorithm weights are equal. M2 also has a censored form, in which failures are right-censored at the budget rather than assigned a fictitious expansion count. M3 counts transitions along the selected goal state’s trusted predecessor chain and is undefined, never imputed, for unsolved episodes. M4 measures agreement with the current additive heap head over decisions with at least two live choices, against a chance rate equal to the decision-weighted mean of $1/K$. M5 restricts to tasks where the mean $2 \times$ random-valid solve indicator lies strictly between 0 and 1. Because goal selection is the (R_t+1) -th decision, the exact reference fails at $m = 1$ and scores $M1 = 0.875$ by construction, and arms that reach the goal in fewer decisions can exceed it (privileged hadd-greedy 0.882, 0.928, and 0.906).

J RESULTS DETAIL: GATE RECEIPTS AND QUALIFICATION GATES

This appendix gives the program’s receipts, the follow-up ledgers, and the qualification gates that Section 4 summarizes.

Table 6: Frozen 15-task text-state BFWS development panel, with full-episode budgeted success by condition. The learned-to-best-control gain over random-valid is 0.467 with a paired whole-instance bootstrap 95% interval lower bound 0.2 (10,000 resamples). Single fixed training seed, development stage, final-evaluation manifest not accessed.

Condition	Success (15 tasks)
Pretrained base	0.0
Process SFT	0.933
Random-valid (oracle-assisted control)	0.467
Exact execution	1.0

The expanded study program ran on 2 A100 80GB GPUs under a 336 GPU-hour cap with an absolute scheduler cutoff. It spent 53.52 of those hours (15.9%) at the synthesis snapshot, before the corruption and seed-replication addenda, and 68.5244 after them. Seven branches were admitted. Five completed at v1 scope, and two (second backbone, generalization/robustness) stopped at frozen v1 cost-admission gates with terminal VALID_STOP receipts, then completed at reduced v2 scope under new versioned protocols. Table 3 combines frozen scope, status, coverage, and ledger spend. Every executed branch reconciles with zero missing evidence, with 1,152/1,152 baseline episodes independently replayed, 405/405 DAGger episodes, and 90/90 successor-prediction episodes.

Follow-up windows executed after the program on separate GPU ledgers (Table 3) and on CPU only (the comparator zoo, the counterfactual audit, and the first external audit). These windows are reported separately and never summed with the main program ledger. The native-arms window spent 3.2086 of 12 GPU-hours, the R1-R4 follow-up window spent 7.7998 of 36, and the choice-frontier window spent 1.069 of 12. The R1-R4 follow-up window was resized from 20 to 36 GPU-hours before any outcome, because its frozen worst-case estimate of 27.04 GPU-hours including spend exceeded the initial cap, and realized cost landed at 29% of that estimate. Each follow-up closed with complete coverage and independent replay (Table 3).

Of three earlier gates that frame the active algorithm matrix, the bounded BFS pilot closed with a governed VALID_STOP. On a frozen outcome-blind 15-task BFS development panel, final checkpoints from five process-SFT seeds achieved 1.0 invariant-valid success with a 0.0 invalid-operation rate. The base achieved 0.0 and random-valid achieved 1.0, so the learned policy showed executable BFS behavior with zero gain over the strongest control, and downstream work was gated off. The BFWS structural gate then passed on its frozen development panel (Section 5.1). The overlap between the BFWS and BFS development panels is not pinned. The companion manifest of the frozen 45-task final evaluation was fixed before outcomes and has never been accessed.

The optimal A* arm over $h_{\max}$ with landmark counting was retired after its own governed VALID_STOP and is superseded by the two additive best-first variants. Its original heuristic-representation estimand remains unidentified. The historical five-seed BFS pilot predates the budget rule, which authorizes one training run at seed 17 per enumeration-contract model and cell (Section 4).

K RESULTS DETAIL: PRIMARY-MATRIX TABLES

Table 6 gives the BFWS development panel and Table 7 the expanded 24-task baseline by algorithm and modality, both summarized in Section 5.1. Measures are reported separately and never combined.

L RESULTS DETAIL: PRIMARY-MATRIX PAIRED CONTRASTS

This appendix gives the paired whole-problem contrasts of the expanded baseline, for readers asking how the learned gain compares with each control.

Paired whole-problem contrasts supply the learned-to-best-control gain, kept separate from the success counts (Table 8). Against the base, process SFT gains only in the additive cells (+0.833 for weighted A*, +0.903 for greedy) and gains nothing under BFS or BFWS. Against random-valid, process SFT loses in every algorithm column and wins zero of the 72 problem-modality units in any column. Random-valid itself ties exact execution in both additive columns, so those cells cannot

Table 7: Primary matrix of the expanded 24-task baseline by algorithm and modality. Measures are reported separately and never combined. Columns give full-episode budgeted success out of 24 for each condition, process-SFT invalid operations over decisions (inv/dec), process-SFT expansions, and exact-reference decisions as the budget reference. The pretrained base terminates on its first invalid operation in every cell (24/24 inv/dec, 0 expansions). Random-valid (RV) is an oracle-assisted programmatic control and exact execution is the oracle upper bound, and both are references rather than model abilities.

Algorithm	Modality	Base	SFT	RV	Exact	SFT inv/dec	SFT exp.	Exact dec.
BFS	text	0/24	0/24	15/24	24/24	24/66	15	1969
BFS	visual	0/24	0/24	15/24	24/24	24/69	17	1969
BFS	multimodal	0/24	0/24	15/24	24/24	24/63	15	1969
BFWS	text	0/24	0/24	17/24	24/24	24/24	0	974
BFWS	visual	0/24	0/24	17/24	24/24	24/24	0	974
BFWS	multimodal	0/24	0/24	17/24	24/24	24/24	0	974
Weighted A* ($w=3$)	text	0/24	20/24	24/24	24/24	4/609	179	694
Weighted A* ($w=3$)	visual	0/24	18/24	24/24	24/24	6/499	143	694
Weighted A* ($w=3$)	multimodal	0/24	22/24	24/24	24/24	2/666	196	694
Greedy best-first	text	0/24	22/24	24/24	24/24	2/614	179	673
Greedy best-first	visual	0/24	22/24	24/24	24/24	2/643	189	673
Greedy best-first	multimodal	0/24	21/24	24/24	24/24	3/617	179	673

Table 8: Paired whole-problem success contrasts on the expanded baseline (10,000 paired resamples over the 24 problems, seed 1729). Entries give the point contrast with wins/losses/ties over the 72 problem-modality units. Contrasts are descriptive bounds over a tiny subgroup and a single training seed (Efron, 1979).

Contrast	BFS	BFWS	Weighted A* ($w=3$)	Greedy
SFT – base	+0.000 (0/0/72)	+0.000 (0/0/72)	+0.833 (60/0/12)	+0.903 (65/0/7)
SFT – random-valid	−0.625 (0/45/27)	−0.708 (0/51/21)	−0.167 (0/12/60)	−0.097 (0/7/65)
SFT – exact	−1.000 (0/72/0)	−1.000 (0/72/0)	−0.167 (0/12/60)	−0.097 (0/7/65)
Random-valid – exact	−0.375 (0/27/45)	−0.292 (0/21/51)	+0.000 (0/0/72)	+0.000 (0/0/72)

measure any advantage beyond valid bookkeeping. These contrasts are tiny-subgroup descriptive bounds over 24 problems and one training seed. They do not support modality or algorithm superiority claims beyond the executed panel. The development-panel BFWS positive does not reappear on this suite. The complete six-measure matrix for the frozen final protocol, including per-cell Algorithm Invariant compliance and charged-budget columns, belongs to the unexecuted final evaluation.

M RESULTS DETAIL: SECONDARY PANELS

This appendix reports the secondary enumeration-contract panels, which Section 5.1 points to and which sit outside the choice-quality claim.

Modality. The per-modality rows of Table 7 show the same qualitative profile in all three adapters, with zero learned success under BFS and BFWS and learned success below the saturating controls in the additive cells. The earlier matched-modality study is a 3-problem panel on its own ledger that is never pooled with the expanded baseline. It gives process-SFT success of 5/12 in text-state, 6/12 in visual-state, and 6/12 in multimodal-state, versus 10/12 for random-valid in each modality, 0/12 for the base, and 12/12 for exact execution. The evidence does not support an intrinsic modality effect. Two caveats bound any such claim. The unlabelled 128px rendered views can lose symbolic identities, and the static and goal pages still contain text in every modality, so semantic parity across adapters is not established.

Generalization and robustness v2. The reduced v2 protocol admitted a uniform cheapest-prefix sample of 5 variants per family over 5 perturbation families (25 of 93 eligible semantics-preserving variants, membership frozen before execution) and evaluated the learned additive-best-first adapters. Learned adapters succeeded on 128/150 episodes versus 0/150 for the pretrained base, where every base episode is a one-call invalid termination (150 decisions total). By family, learned success was 29/30 on object-renaming, 27/30 on name-compression, 27/30 on render-restyle, 26/30 on scale-up, and 19/30 on shifted-init. By modality the split was text 44/50, multimodal 43/50, visual 41/50. By

Table 9: DAgger on BFS, with development and unseen success and unseen invalid-operation rates, against the exposure-matched continued-SFT control and the two references.

Arm	Dev success /9	Unseen success /72	Unseen invalid rate
DAgger iteration 2	3/9	1/72	0.316
Continued SFT iteration 2	2/9	1/72	0.175
Original process SFT	1/9	0/72	0.364
Random-valid	6/9	45/72	0.000
Exact reference	9/9	72/72	0.000

algorithm it was greedy 67/75 and weighted A* 61/75. All 22 learned failures are single-invalid-operation terminations, 11 of them in shifted-init. Controls saturate (random-valid 750/750, exact 150/150) and remain bounds, not learned ability. The lossy name-compression stratum is reported separately and never pooled. All 5 admitted variants are text-lossy, 3 multimodal-lossy, and 0 visual-lossy, with learned success on that stratum of text 9/10, visual 8/10, multimodal 10/10. Family strata hold 5 problems each, so family-level rates are descriptive only. The full-scope v1 generalization branch was withdrawn after its terminal VALID_STOP receipt (closed 2026-09-21) and will not be executed.

DAgger. Two DAgger iterations (Ross et al., 2011) on BFS were compared against an exposure-matched continued-SFT control on 3 development and 24 unseen modality-task pairs (Table 9). On the unseen panel, DAgger and continued SFT each achieve 1/72 invariant-valid successes (paired over whole problems, one DAgger win, one loss, 70 ties). Both are far below random-valid at 45/72 and exact execution at 72/72, while the original process-SFT checkpoint achieves 0/72. The result is null. DAgger also carries the higher unseen invalid-operation rate of the two iterated arms (0.316 versus 0.175). Final cumulative unique training corrections were text 201, visual 203, multimodal 181, accumulated over 12 training cells of 512 records and 16 optimizer updates per cell-iteration.

Model-generated successors. Replacing trusted successors with model-generated dynamic-state successors collapses goal success from 45/45 to 1/45 on the evaluated panel, with zero trusted-state substitutions in the trusted arm. The single success is the one multimodal unseen cell. The trained successor model is exactly correct on 65/109 retained raw predictions, with 93 schema-valid and 66 effect-valid, and downstream failure kinds are dominated by exact state-identity and effect failures (27 effect, 16 schema, 1 static-context) rather than schema failures. Collection-time verification of the untrained exact-successor accepted 119/1,536 predictions (text 43, visual 52, multimodal 24). Coverage is partial by design. The frozen outcome-blind admission accepted a 12-task fallback panel after the full 24-task unseen panel was priced at 147.66 GPU-hours against the 64 GPU-hour branch cap, so full unseen coverage remains missing.

Curriculum by modality. Nine training cells (three modalities under staged, shuffled, and mixed ordering, all under greedy additive best-first and trained fresh from the same base on identical record sets) were evaluated on 27 unseen whole problems. Successes were 24/24/25 (staged/shuffled/mixed) in text-state, 24/27/26 in visual-state, and 25/27/27 in multimodal-state. A sequential-SFT comparator reached 24/27, 25/27, and 24/27 in text, visual, and multimodal cells respectively, the base stayed at 0/27, and random-valid and exact execution saturated at 27/27 in every modality. All four paired interaction intervals include zero: $-0.111 [-0.259, +0.037]$ and $-0.074 [-0.259, +0.074]$ for the visual and multimodal staged-minus-shuffled contrasts against text, and $-0.074 [-0.259, +0.111]$ and $-0.037 [-0.185, +0.111]$ for the corresponding mixed-minus-shuffled contrasts. No material ordering-by-modality interaction emerged. Because both references saturate and the base sits at floor, ceiling and floor effects bound the observable differences.

Transfer. All 12 verified planning adapters plus the base were evaluated zero-shot on frozen subsets of FOLIO (Han et al., 2022) (200 items), GSM8K (Cobbe et al., 2021) (200), and HumanEval (Chen et al., 2021) (164). Table 10 gives the per-adapter accuracies. The paired verdict uses McNemar exact comparisons (McNemar, 1947) of each adapter against the per-example base predictions, Holm-corrected across all 36 comparisons (Holm, 1979). None survives correction. The minimum adjusted p is 0.8156, and the smallest raw p is 0.0227 (FOLIO with the text width-based adapter, at +12 examples). Leakage screening over the frozen 8-gram window finds 7/564 benchmark items

Table 10: Zero-shot transfer accuracy on frozen benchmark subsets (FOLIO 200, GSM8K 200, HumanEval 164). Adapter rows follow the transfer ledger names: `bfs`, `iw` (width-based best-first, BFS), `astar_w3`, and `astar_greedy`, each in text (t), visual (v), and multimodal (m) training cells. No comparison against the base survives Holm correction (0/36, minimum adjusted p 0.8156).

Adapter	FOLIO	GSM8K	HumanEval
Base	0.600	0.955	0.780
<code>bfs t/v/m</code>	0.600 / 0.605 / 0.605	0.950 / 0.945 / 0.965	0.793 / 0.799 / 0.780
<code>iw t/v/m</code>	0.660 / 0.645 / 0.645	0.960 / 0.965 / 0.960	0.768 / 0.762 / 0.756
<code>astar_w3 t/v/m</code>	0.590 / 0.575 / 0.590	0.950 / 0.965 / 0.945	0.817 / 0.799 / 0.774
<code>astar_greedy t/v/m</code>	0.590 / 0.590 / 0.590	0.950 / 0.955 / 0.950	0.799 / 0.811 / 0.793

sharing only trivial numeric 8-grams with the planning corpus and zero content overlap. The transfer result is a null.

Second backbone. Two reduced-scope cells trained and evaluated InternVL3.5-8B (Wang et al., 2025). Each used a frozen 12-task key-cell panel selected by lowest reference decision count before any model outcome, with one seed-17 training run per modality cell. The earlier cell (protocol v2) ran BFS, and neither arm reached a goal. The pretrained base scored 0/36 at 1 decision per episode (36 total), process SFT scored 0/36 at 1 to 4 decisions (59 total), and the reused BFS controls scored random-valid 54/72 (18/24 per modality, oracle-assisted) and exact 72/72. The replication cell (protocol v3, stress test R3) ran greedy additive best-first with the same backbone, recipe, and panel shape. Process SFT succeeds on 11/12 text-state, 10/12 visual-state, and 10/12 multimodal-state problems (31/36), while the base scores 0/36 with one invalid call per episode. Process-SFT invalid-operation rates are 0.00546, 0.01220, and 0.01258, and decision usage is 183, 164, and 159 of a 390-call allowance (text, visual, multimodal). Random-valid and exact reference both score 24/24 per modality. These control counts hold two entries per panel task, and both entries of a task come from its one reused, hash-pinned greedy baseline control episode per condition and modality. Paired whole-problem contrasts (seed 1729, 10,000 resamples) give SFT minus base $+0.917$ [$+0.750, +1.000$] in text-state and $+0.833$ [$+0.583, +1.000$] in both visual-state and multimodal-state. SFT minus random-valid is -0.083 [$-0.250, +0.000$], -0.167 [$-0.417, +0.000$], and -0.167 [$-0.417, +0.000$] in the same order. Coverage is complete, with 72/72 model episodes and 72/72 comparators replayed. Learned success under the enumeration contract appears only in the additive cells, where the identity audit shows no decision to make (Section 5.1), and the InternVL v3 cell replicates that contract fluency (31/36 versus 0/36 base, below random-valid 24/24 per modality). The earlier total null on this backbone is consistent with an algorithm (choice-requirement) difference rather than a backbone null. The cross-backbone contrast against the pinned Qwen3-VL process-SFT rate is descriptive rather than paired, because the InternVL panel is the key-cell subset while the Qwen rate is the pinned full-panel aggregate. It is $+0.000$ [$-0.167, +0.083$] in text-state, -0.083 [$-0.333, +0.083$] in visual-state, and -0.042 [$-0.292, +0.125$] in multimodal-state. InternVL3.5-8B shares the Qwen3-8B LLM family (Yang et al., 2025), so the comparison differs in the vision tower, connector, image tokenization, and multimodal recipe, and it is reported as an architecture-difference limitation. The full-scope v1 second-backbone branch was withdrawn after its terminal VALID_STOP receipt (closed 2026-09-21) and will not be executed, and its gate is retained as history.

N RESULTS DETAIL: FAILURE-MECHANISM CALIBRATION

This appendix localizes where each arm class fails, for readers asking why learned success is absent under BFS and BFS.

A frozen CPU-only calibration mined 2,910 unique replay-verified episodes across the five executed model branches and classified all 905 failures with zero unclassified. The protocol was frozen before any outcomes were computed, and the analysis is descriptive. Three arm-level profiles emerge. The base policy fails in 447 of 447 episodes, always on the first decision. Of these, 379 emit a malformed output that never reaches the operation checker, and 67 issue rejected BFS frontier operations. Learned arms fail in 407 of 906 episodes, and the dominant learned failure kind is `other_invariant_violation` with 229 occurrences (56.3% of learned failures), namely parsed, source-correct, applicable operations that the runtime rejects because they violate the declared search invariant. The remaining learned failures are 72 malformed outputs, 56 inapplicable

Table 11: Observation corruption on frozen checkpoints ($k=9$ admitted tasks, paired whole-problem deltas against clean runs, strata below 8 descriptive-only under the frozen rule). “Material” is the frozen protocol’s interval-based verdict.

Corruption	Arm	Corrupted	Clean	Δ	95% interval	Material
Text-masked	text	0.000	0.944	-0.944	$[-1.000, -0.833]$	yes
Text-masked	multimodal	0.000	0.889	-0.889	$[-1.000, -0.722]$	yes
Text-shuffled	text	0.000	0.944	-0.944	$[-1.000, -0.833]$	yes
Text-shuffled	multimodal	0.167	0.889	-0.722	$[-0.889, -0.500]$	yes
Visual-blank	visual	0.833	0.889	-0.056	$[-0.167, +0.000]$	no
Visual-blank	multimodal	0.778	0.889	-0.111	$[-0.278, +0.000]$	no
Visual-degraded	visual	0.833	0.889	-0.056	$[-0.167, +0.000]$	no
Visual-degraded	multimodal	0.778	0.889	-0.111	$[-0.278, +0.000]$	no

grounded actions, 27 effect failures, 16 schema failures, and smaller kinds. First failures cluster early, at decision indices 0, 1, and 2 with counts 137, 88, and 46. The oracle control fails 51 times, all late and all by expansion-budget exhaustion (buckets 5–8, 9–16, 17–32, and 33+ hold 9, 9, 3, and 30). Table 2 gives the rejection taxonomy.

Every baseline process-SFT episode under BFS or BFWS fails (144/144). The DAGger-iteration BFS arms fail 155/162, while the learned additive best-first families fail 56 of 546 episodes (10.3%). This pattern matches the primary matrix (Section 5.1). The calibration is descriptive. After process supervision, most learned failures (229/407) are parsed, source-correct operations rejected by the search invariant, and 72/407 remain malformed outputs. Read with the horizon profile, the learned policy fails early on invariant grounds, whereas the oracle control, which always emits valid operations, fails only when its expansion budget runs out late in the episode.

O RESULTS DETAIL: OBSERVATION-CORRUPTION CHANNEL TESTS

This appendix reports the observation-corruption tests, which show which observation channel the trained enumeration-contract controller relies on.

The observation-corruption suite evaluated frozen checkpoints, with no retraining, on a prospectively frozen set of $k=9$ admitted tasks under four deterministic corruption families applied at observation time. Clean learned success on these tasks is 0.917 (text-state 0.944, visual-state 0.889, multimodal-state 0.889, with 9 tasks $\times$ 2 algorithms per cell). Table 11 gives the paired whole-problem deltas. Copyability-preserving text shuffling lowers success by up to -0.944 $[-1.000, -0.833]$, and the multimodal arm retains at most 0.167 success under shuffled text. Masking instead destroys the output contract. Visual-channel corruption is not material. Blanking or degrading the rendered state costs at most -0.111 $[-0.278, +0.000]$, and every visual interval includes zero. The multimodal channel-isolation contrast, paired by task and algorithm ($n=18$), is -0.806 for text-channel corruption versus -0.111 for visual-channel corruption, giving text minus visual -0.694 with 95% interval $[-0.889, -0.472]$, which is material under the frozen rule. This contrast pools the output-contract-destruction (masked) regime. Controls behave as the contract predicts. The corrupted base stays at floor (0/144, all first-decision invalid), random-valid succeeds 72/72 on the new frozen seeds, so per the saturation rule no structural-advantage claim is made, and exact execution remains at 18/18.

A decomposition of the 72 published text-corruption episodes (stress test R2) separates two regimes that the success deltas pool together. Under text masking, every episode of the text-state and multimodal-state cells is all-invalid (18/18 in each, 18 invalid operations over 18 decisions). Each emitted operation fails the schema, so the policy never enters the search. This is output-contract destruction, not information loss. Copyability-preserving text shuffling behaves differently, because valid operations keep flowing while the search fails. In the text-state cell 16/18 episodes are partial-valid, 2/18 are all-invalid, and none reaches the goal (18 invalid operations over 49 decisions). In the multimodal-state cell 15/18 are partial-valid, none is all-invalid, and 3/18 reach the goal (15 invalid operations over 105 decisions). This partial-valid pattern is the information-loss signature. A new cross cell applies text masking to the frozen visual-state arm, whose nominal state channel is the image, and it scores 0/18 with every episode all-invalid. Masking the text scaffold destroys the output contract even when images are the nominal state channel.

Table 12: Menu manipulation on the frozen native adapters (both arms, two additive algorithms). Invalid operations are counted over emitted operations. The distractor pick-rate is counted over decisions with injected distractors.

Family	Success	Invalid ops	Distractor picks
Menu permutation	36/36	0/512	n/a
Distractor injection	0/36	36/60	36/60 (0.6)

The language-channel claim survives only in its shuffled form. The masked catastrophe in Table 11 is mostly a loss of the copyable output format rather than a loss of state information. Shuffling the text leaves operations copyable and still stops the search, so under the frozen observation contract the textual serialization carries decision-relevant information for these adapters. Removing or degrading the visual channel barely moves success, including on the visual-state arm whose only rendered-semantics channel was corrupted. Sections P and Q probe the grounded candidate menu directly. This is a channel finding about the trained controller under the declared adapters, not evidence about intrinsic modality value.

P RESULTS DETAIL: MENU-MANIPULATION RECEIPTS

This appendix reports the menu-manipulation receipts, for readers asking whether the trained enumeration-contract policy reads its menu’s content or only its surface.

Menu manipulation (stress test R1) probes how the trained policy responds to changes in the grounded menu. It runs on the frozen image-only adapters of both native arms (the Image-Only Native Arms appendix, Appendix Q), with no retraining, under two families. Menu permutation reorders the candidate menu. Distractor injection adds $k=3$ schema-valid but inapplicable actions to it. Table 12 reports the 72 episodes, all independently replayed within the 170/170 stress-evaluation episodes.

On the frozen native adapters (stress test R1, 72 replayed episodes), menu permutation leaves 36/36 successes and 0/512 invalid operations, which is uninformative about order under the submission-order invariance of Section 5.1. Injecting $k=3$ schema-valid but inapplicable actions collapses success to 0/36, with 36/60 invalid operations and a distractor pick-rate of 0.6 against a uniform-picking expectation of 0.592 to 0.625. Every episode ends on a distractor pick, 20/36 on the first decision. The receipt therefore shows that the policy does not reject schema-valid inapplicable menu entries (pick-rate at chance), and these out-of-distribution distractors never appeared in training menus. Content-reading and position habits both predict a pick-rate near 0.6, so this receipt does not separate them, and the discriminating fixed-versus-random-position test remains unrun.

Q RESULTS DETAIL: IMAGE-ONLY NATIVE ARMS

This appendix reports the image-only native arms, which test whether removing the textual scaffold changes the enumeration-contract results.

The native-arms follow-up (contract v2) trained two image-only observation arms. In `visual-nomem-state` (nomem) and `visual-seq-state` (seq), state facts arrive only as rendered images, and seq adds a window of $K=8$ history frames. Neither arm receives textual search memory, g/h /priority scalars, candidate scores, or membership flags. The grounded, unscored candidate menu is still presented. The v1 admission gate measured pretrained-base format compliance and is retained as history. The v2 amendment, frozen before any trained-arm outcome, replaced it with a pre-registered post-training smoke gate, which both arms passed with 62/62 schema-valid grounded calls against a threshold of 0.5. Four adapters (two arms by two additive best-first algorithms) used the frozen recipe with 512 records, 16 updates, and seed 17. All 216/216 new bindings were recorded and independently replayed, together with 72/72 reused replay-verified comparators. The window spent 3.2086 of 12 GPU-hours.

On the clean panel, both arms nominally succeed on 18/18 with zero invalid operations, against 16/18 for the published frozen `visual-state` arm. The pretrained base scores 0/36 under both new contracts and exact reference scores 18/18. Random-valid is at ceiling on its five frozen seeds, so under the saturation rule no structural-advantage claim is made. Paired whole-problem con-

Table 13: First measurement under the choice-frontier contract on the frozen 9-task panel, with 18 cells (9 tasks $\times$ 2 additive algorithms as repeated measures) and 144 episodes independently replayed. Exact reference solves by construction of the budget and is the imitation target. Not comparable with enumeration-contract rows.

Condition	Goal rate	Failures	Invalid ops
Exact reference	18/18	none	0
Learned adapter	7/18	11 budget exhausted	0
Pretrained base	0/18	18 invalid at first call	18
Random-valid (5 frozen seeds)	32/90	58 budget exhausted	0

trasts ($n=18$, bootstrap seed 61813, 10,000 resamples, 95%) give `nomem minus visual-state` $+0.111$ $[0.000, +0.278]$ and `seq minus visual-state` $+0.111$ $[0.000, +0.278]$, neither material. The history window changes nothing on these tasks (`seq minus nomem` 0.000 $[0.000, 0.000]$). Learned minus pretrained base is $+1.000$ $[+1.000, +1.000]$ for each arm, which is material.

The pre-registered corruption payoff is a null. Under `seq` the images are the only state and history channel, yet blanking every attached image gives a `seq corrupted-minus-clean` contrast of 0.000 $[0.000, 0.000]$, and degrading every image gives 0.000 $[0.000, 0.000]$. The images-only hypothesis is therefore answered in the negative on this panel. As a post-hoc interpretation, the grounded unscored candidate menu is the remaining channel that could carry the signal, since neither the rendered scenes nor the removed scalar scaffold is necessary once the menu is present. On these additive cells the identity audit (Section 5.1) makes any listed candidate a correct emission, so the null is equally consistent with a policy that uses no state information.

The degraded-strain test (stress test R4) evaluated the same frozen native adapters on the two generalization families where the full-scaffold adapters had degraded. The learned adapters succeed on 10/10 shifted-init and 10/10 scale-up episodes for both arms, and the pretrained base scores 0/20 on each strain (0/40 in total). The published full-scaffold rates on the same variants are 19/30 for shifted-init and 26/30 for scale-up (Secondary Panels appendix, Appendix M). The two rates come from different episode sets and are not paired, and the reduced contract sits at ceiling where the full scaffold degraded. A post-hoc, unpaired reading, with 10 visual-only episodes per arm against 30 full-scaffold episodes pooled over three modalities, is consistent with the reduced contract removing scalar operands whose emission the full-scaffold adapters failed. It does not measure whether that text carried signal, since any listed candidate succeeds on these cells.

R RESULTS DETAIL: CHOICE-FRONTIER AND COMPARATOR-ZOO RECEIPTS

This appendix holds the receipts of the first choice-frontier measurement and the comparator zoo on the reused panel, for readers checking the contract’s first identity-audit pass.

The training corpus was derived on CPU by replaying the frozen exact traces through the new contract and verified expansion-for-expansion against them (58 episodes bit-exact). It covers 29 training tasks and holds 1,046 records (512 training plus 11 diagnostic per algorithm) with 3,544 scenes. Every corpus episode carries at least one decision whose menu offers two or more states (up to 315 such decisions in one episode, menus of up to 222 states). Two adapters, one per additive algorithm, were trained with the frozen recipe at seed 17 (16 updates, 512 records each), and the smoke gate accepted 40/40 model calls. All 144/144 panel episodes were independently replayed with complete coverage. The window spent 1.069 of 12 GPU-hours (training 0.81, smoke 0.052, model evaluation 0.208), and corpus and controls ran on CPU.

Paired bootstrap contrasts (seed 61813, 10,000 resamples) give `random-valid minus exact reference` -0.644 $[-0.856, -0.422]$, material, so the controls separate under this contract. The learned adapter minus pretrained base is $+0.389$ $[+0.167, +0.611]$, material, with zero invalid operations, a contract-fluency gain. The learned adapter minus random-valid is $+0.033$ $[-0.033, +0.133]$, not material. The 18 cells are 9 tasks with two algorithms as repeated measures, and that last contrast rests entirely on one task (storage), the only one where random-valid’s seed frequency lies strictly between 0 and 1. That falls below the frozen rule that strata under eight cells are descriptive only, so the contrast is descriptive. The evidence supports no learning-to-plan claim for the adapter, and no equivalence claim with random-valid, since no equivalence margin was pre-registered.

Table 14: Exploratory pre-registered metric set at the frozen shared $2\times$ cap (9 panel tasks $\times$ 2 additive settings). M1 is the task-clustered solve-versus-budget AUC over $m \in \{1, 1.25, 1.5, 1.75, 2\}$ (9-task bootstrap, seed 133, 10,000 draws). M2 is the solved-only task-weighted geometric expansion overhead ρ and M3 the solved-only task-weighted cost ratio κ , both selection-conditioned. The $3\times$ and $4\times$ columns are additional CPU controls, not M1. Selectors read privileged information that the model does not see.

Arm	M1 AUC [95% CI]	$2\times$ solved	$3\times$	$4\times$	M2 ρ	M3 κ
Learned adapter	.1042 [.0278, .2014]	7/18	n/a	n/a	1.432	1.025
Pretrained base	0 [0, 0]	0/18	n/a	n/a	undefined	undefined
Random-valid	.1389 [.0250, .2722]	32/90	42/90	50/90	1.407	1.000
Exact reference	.8472 [.7917, .8750]	18/18	n/a	n/a	1.000	1.000
bfs-order	.0694 [0, .1528]	6/18	8/18	10/18	1.613	1.000
Novelty-first selection	.1250 [0, .2778]	6/18	8/18	12/18	1.452	1.000
worst-first	.0556 [0, .1389]	4/18	6/18	8/18	1.549	1.050

The control expansion sequences coincide across the two algorithms on every panel task. Post-hoc task-clustered intervals with the same seed and resample count are $[0.000, +0.100]$ for learned minus random-valid and $[-0.889, -0.333]$ for random-valid minus exact reference.

Comparator zoo. The CPU-only comparator zoo evaluates this measurement against three deterministic selectors at the same $2\times$ cap (bfs-order, novelty-first selection, and negative control worst-first). Its metric set was pre-registered before comparator episodes, but stored choice-frontier episodes enter post-hoc. The selectors act on the additive frontier, not native BFS or BFWS, and read privileged scalars or PDDL atoms while the model sees only scenes, so no information-matched learned-versus-rule claim follows.

On M1 the learned adapter falls between bfs-order and random-valid and below novelty-first selection (Table 14, Figure 3). No comparator between random-valid and exact exists on this panel, so the zoo bounds the adapter only against weaker-than-random rules and cannot calibrate choice quality. Post-hoc M4 finds 17.4% heap-head agreement against 26.9% chance and 54.5% last-menu-label emissions, and the task-clustered interval of the agreement gap includes zero (Counterfactual appendix, Appendix S), with no last-label comparator and no equivalence claim. Across the fifteen-test Holm family no comparison supports positive learned Pareto dominance, with three co-solved tests undefined. Non-significance is neither equivalence nor a generalized negative capability claim, and the one-task M5 subset is descriptive only.

Zoo receipts. Random-valid is the zoo’s uniform-choice reference and exact reference the imitation target. Random-valid and the three selectors also ran at $3\times$ and $4\times$ caps, outside the primary metric. Because the CPU selectors operate on the additive frontier, novelty-first selection does not reproduce BFWS pruning. Selector episodes replay the frozen model input, randomized menu, actions, admission, and budget, but do not generate or replay image view bindings. All 432/432 zoo bindings were observed and independently replayed, with 342 new CPU episodes, 90 original $2\times$ random-valid controls reused, and none missing. The nine unit-cost optimal costs C^* come from one offline planner computation per task with replay-verified plans: storage 3; elevators, ferry, and 15puzzle-compact 4; depot, towers-of-hanoi, and visitall 5; blocksworld and 15puzzle-expanded 6.

The selectors are bfs-order (minimum frontier g , then earliest serial), novelty-first selection (lowest width-1 then width-2 novelty over positive PDDL atoms), and worst-first as a negative control.

Exploratory post-hoc M4 on the stored episodes shows the learned adapter selecting the teacher’s heap-head state on 23/132 multi-option decisions (17.4%) against an on-policy chance rate of 26.9%, and emitting the last menu label on 72/132 (54.5%). Decisions cluster within 9 tasks, and the task-clustered intervals in the Counterfactual appendix (Appendix S) show that the agreement gap does not survive that clustering while the last-label excess does. The post-hoc choice-frontier diagnostic records a first-label rate of 2.3% and a teacher-corpus last-label rate of 12.0% (chance 10.4%, 978 records). Menus are permuted per decision with a seed, and no last-label comparator was run, so these post-hoc rates do not establish a position habit. M5 restricts to tasks where random-valid solves at a $2\times$ rate strictly between 0 and 1. That subset holds one task (storage-compact-919000, random-valid rate .2), so M5 is descriptive only. On it the learned adapter’s M1 AUC is .3125, against .075 for random-valid and .625 for exact reference.

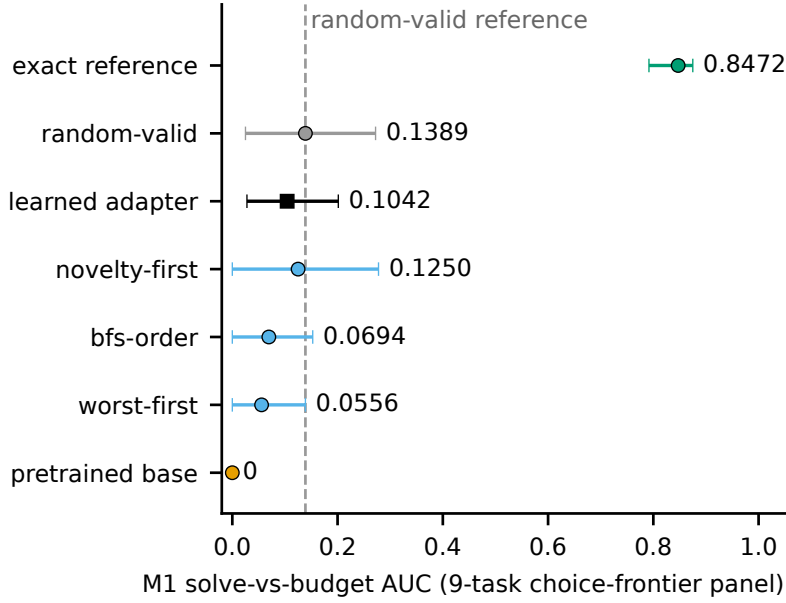


Figure 3: Solve-versus-budget AUC (M1) on the frozen 9-task choice-frontier panel at multipliers 1 to 2. Exact reference .8472, random-valid uniform-choice reference .1389, learned adapter .1042 [.0278, .2014], novelty-first .1250, bfs-order .0694, worst-first .0556, pretrained base 0. Intervals of all non-exact arms overlap and the Holm family licenses no ordering. Under the enumeration contract, the BFS/BFWS random-valid rows (15/24 and 17/24 versus exact 24/24) already registered choice.

Table 15: Random-valid against exact reference on the 48 additive pairs under the two heap-serial rules.

Heap-serial rule	Identical	Divergent
Sorted candidate order (executed)	48	0
Submission order (counterfactual)	29	19

The choice-frontier contrasts use percentile 95% intervals with the frozen bootstrap resampling cells. Random-valid’s seed frequency on storage is 0.2. The final audit returns CHOICE_SENSITIVE and passes over 144 replayed episodes.

S RESULTS DETAIL: SUBMISSION-ORDER COUNTERFACTUAL AND M4 INTERVALS

This appendix gives the submission-order counterfactual behind the enumeration-contract identity audit and the task-clustered intervals for the first adapter’s post-hoc diagnostics.

A CPU-only follow-up with a protocol committed before its outputs re-ran the identity audit on the same 48 additive pairs (24 tasks $\times$ 2 algorithms, seed 17). The executed rule assigns heap serials at expansion start from the sorted candidate order, and the counterfactual assigns them at application time in submission order. The audit first re-validated 288 stored episode counts and 96 raw-output sequences with no mismatch.

The stored verdict reads “PREDICTION_FAILED: see preregistered_prediction”. The executed rule stays identical as predicted, but 19/48 divergent pairs fall short of the pre-registered threshold of more than 24/48. First divergence falls between decisions 2 and 50 (median 6), and both algorithms on one ferry task flip the goal outcome. The 48/48 identity is therefore a property of the executed serial rule, not of the enumeration contract alone.

A post-hoc analysis recomputed the first adapter’s M4 diagnostics over its 18 stored episodes on the earlier panel (9 task clusters, 10,000 resamples, seed 133). Pooled over 132 multi-option decisions, heap-head agreement is 0.174 against a chance rate of 0.269, and the last-label rate is 0.545. The

Table 16: Post-hoc M4 diagnostics for the first adapter on the earlier 9-task panel.

Quantity	Pooled	Task-clustered [95% CI]
Heap-head agreement minus chance	-0.095	-0.069 [-0.161, +0.032]
Last-label rate minus chance	+0.276	+0.277 [+0.126, +0.402]

Table 17: The frozen validation panel (12 tasks, 7 domains). R is the uncapped exact-reference expansion count (greedy / w3), and Screen counts random-valid goals over 10 screening episodes.

Task	C^*	R	Screen
15puzzle-935006	8	46 / 12	1
blocksworld-935001	10	26 / 26	4
blocksworld-935004	10	13 / 13	2
depot-935005	10	14 / 15	1
depot-935011	9	13 / 13	1
elevators-935000	8	11 / 11	2
elevators-935013	8	11 / 11	2
grid-935016	8	9 / 9	4
storage-935000	10	20 / 17	1
storage-935001	8	18 / 15	2
towers_of_hanoi-935000	13	24 / 25	1
towers_of_hanoi-935002	15	24 / 26	2

agreement gap’s interval includes zero while the last-label excess excludes it (Table 16), and M4 keeps its diagnostic, non-confirmatory status.

T RESULTS DETAIL: CHOICE-FRONTIER VALIDATION PANEL AND LADDER

This appendix holds the validation panel’s construction, every arm’s receipts, and all adjacent ladder gaps behind the ladder test in Section 5.2.

The ladder protocol was committed before task generation and the panel membership before evaluation. Of 240 generated candidates (12 snapshots, seeds 935000 to 935019), 127 met the cost filter ($8 \leq C^* \leq 20$) and 34 passed controls-only screening, which admits tasks on which random-valid is discriminative. The freeze rule sorts admitted tasks by domain and seed, keeps at most two per domain, and takes the first 12 (Table 17).

All 696/696 CPU episodes and the 48 first-adapter model episodes were independently replayed. PASS requires all four adjacent paired task-cluster lower bounds to be positive, with arms averaged over seeds within each task and algorithm, then over algorithms. The gaps are +0.082 [+0.051, +0.111] (exact reference minus 0.25), +0.190 [+0.126, +0.252] (0.25 minus 0.50), +0.256 [+0.184, +0.339] (0.50 minus 0.75), and +0.326 [+0.233, +0.422] (0.75 minus random-valid).

Three deviations are documented. Generation reuses the frozen candidate generator, which reproduces all 12 snapshots byte-for-byte although domain generators take the seed differently. A first concurrent generation run corrupted the blocksworld outputs through a shared temporary file, so it was discarded, all candidates were regenerated serially and reproduced, and the deviation was disclosed before evaluation. Only the two additive native references seed the pre-rendered views, rather than four algorithms, because native BFS exceeds the 128-expansion reference ceiling on these tasks, and unseen states are rendered live.

U RESULTS DETAIL: SCALED ADAPTER AND SEED REPLICATION

This appendix gives the adapter’s protocol, corpus, recipe, and per-seed results behind Section 5.3, with the held-out panels, the DAGger ablation, and the zero-shot policy as subsections.

The protocol was frozen before task generation, the corpus before training, and the schedule before launch. D is the mean over training seeds of learned M1 minus random-valid M1, with a task-cluster interval (10,000 draws, seed 133). POSITIVE requires a positive lower bound, $D \geq 0.05$, and a positive estimate at every seed, and the equivalence margin is ± 0.05 . Controls are reused from the ladder test.

Of 2,400 serially generated training tasks, 1,182 remained eligible after excluding overlap with every evaluation panel and deduplicating, and the seed-major walk reached 2048 records per algorithm

Table 18: All arms on the validation panel. M1 carries a task-clustered 95% interval (12-task bootstrap, seed 133, 10,000 draws), and M2 and M3 are solved-only. Only the five ladder arms enter the pre-registered test. The descriptive selectors read privileged information, and the base emits no valid first choice under its one-call cap.

Arm	M1 [95% CI]	$2\times$	M2 ρ	M3 κ
Exact reference	0.875 [0.875, 0.875]	1	1.0	1.021
Exact- ϵ 0.25	0.793 [0.764, 0.824]	0.992	1.089	1.024
Exact- ϵ 0.50	0.603 [0.528, 0.670]	0.95	1.246	1.032
Exact- ϵ 0.75	0.347 [0.252, 0.445]	0.717	1.374	1.023
Random-valid	0.021 [0.007, 0.038]	0.083	1.692	1.042
hadd-greedy	0.882 [0.774, 0.955]	0.958	0.962	1.083
bfs-order	0.000 [0.000, 0.000]	0	undefined	undefined
Novelty-first selection	0.021 [0.000, 0.052]	0.167	1.909	1.000
worst-first	0.000 [0.000, 0.000]	0	undefined	undefined
First adapter	0.026 [0.010, 0.042]	0.208	1.810	1.050
Scaled adapter (seed 17)	0.349 [0.172, 0.531]	0.583	1.326	1.140
Pretrained base	0.000 [0.000, 0.000]	0	undefined	undefined

Table 19: Per-task D (adapter M1 minus random-valid M1) for seed 17 on the validation panel. The pre-registered three-seed endpoint is $D_3 = +0.285 [+0.140, +0.433]$ (Table 21).

Task	D	Task	D
15puzzle-935006	-0.0625	elevators-935013	0.6875
blocksworld-935001	0.0	grid-935016	0.3625
blocksworld-935004	0.0	storage-935000	0.7375
depot-935005	0.0625	storage-935001	0.7
depot-935011	0.7125	towers_of_hanoi-935000	0.2375
elevators-935000	0.5	towers_of_hanoi-935002	0.0

from 236 tasks in 12 domains. Two reseeded menu orders per record give 8,192 samples, with the target last in 17.58% of samples against a chance rate of 17.58%.

The recipe matches the first adapter’s except for corpus, augmentation, sample order, and seed (LoRA rank 64, alpha 128, 128 steps, seed 17). The smoke gate accepted 34/34 calls, and all 24/24 learned episodes, capped at $2R$, replayed with no mismatch and no invalid operation. For seed 17, the $2\times$ solve fraction is 0.583, M2 ρ 1.326, and M3 κ 1.140. Over 691 multi-option decisions heap-head agreement is 0.210 against chance 0.125, the last-label rate is 0.205, and the first-label rate is 0.090.

Four choices are documented. The scope edit to one training seed left seed variance unmeasured in that run. A later pre-registered replication trained seeds 29 and 71 with the byte-identical recipe and the seed-17 sample order, so seeds change only the LoRA initialisation and dropout, and all 48 of its new episodes were independently replayed with no mismatch. The step budget matches the first adapter’s recipe. The seed-major corpus walk resolves an ambiguity in the protocol text. Both evaluation hooks first failed because the scheduler passes only the terminal path, and after a fix deriving the attempt directory from it, the audit re-ran on the same attempts and passed without affecting GPU work or episodes.

On the validation panel, D is $+0.328 [+0.155, +0.508]$ for seed 17, $+0.229 [+0.108, +0.354]$ for seed 29, and $+0.297 [+0.134, +0.466]$ for seed 71. Per-seed M1 is 0.349, 0.250, and 0.318, with a between-seed standard deviation of 0.051 (descriptive). The last-label rate per seed is 0.205, 0.192, and 0.171 (chance about 0.13, descriptive), against 0.300 (chance 0.144) for the first adapter on the same panel. The pre-registered separation from the exact- ϵ 0.75 rung is $S = +0.002, -0.097$, and -0.029 per seed. Against the 0.50 rung the three-seed S is $-0.298 [-0.468, -0.129]$ (descriptive). Table 21 sets these values beside the held-out and unscreened panels.

U.1 HELD-OUT PANELS

This subsection gives the construction, seed amendment, and per-arm receipts of the held-out panel (P2 in tables) and the unscreened panel (P2u), for readers checking how both fresh panels were frozen.

Both fresh panels come from 2,400 candidates generated from seeds 955000–955199. Overlap with earlier panels removed 1,168 candidates, duplicates 205, goal at the initial state 31, and C^* out of range 536, which left 460. The held-out panel applies the validation panel’s generation, C^* , and screening rules. A candidate is kept when $8 \leq C^* \leq 20$ and random-valid reaches the goal in 1 to 9

Table 20: M1 with task-clustered 95% intervals on the validation panel (12 tasks), the held-out panel P2 (11 tasks), and the unscreened panel P2u (12 tasks). All entries are descriptive. Ladder rungs and hadd-greedy read privileged information that no policy sees.

Arm	Validation (12)	P2 (11)	P2u (12)
Exact reference	0.875 [0.875, 0.875]	0.875 [0.875, 0.875]	0.875 [0.875, 0.875]
Exact- ϵ 0.25	0.793 [0.764, 0.824]	0.745 [0.685, 0.805]	0.678 [0.632, 0.721]
Exact- ϵ 0.50	0.603 [0.528, 0.670]	0.507 [0.445, 0.566]	0.397 [0.331, 0.478]
Exact- ϵ 0.75	0.347 [0.252, 0.445]	0.205 [0.145, 0.269]	0.149 [0.091, 0.227]
Random-valid	0.021 [0.007, 0.038]	0.016 [0.000, 0.048]	0.002 [0.000, 0.006]
Pretrained base	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
Adapter, seed 17	0.349 [0.172, 0.531]	0.506 [0.313, 0.699]	0.198 [0.031, 0.396]
Adapter, seed 29	0.250 [0.125, 0.375]	0.284 [0.119, 0.489]	0.120 [0.026, 0.224]
Adapter, seed 71	0.318 [0.151, 0.490]	0.506 [0.295, 0.705]	0.135 [0.031, 0.250]
Adapter, three-seed mean	0.306 [0.158, 0.455]	0.432 [0.273, 0.602]	0.151 [0.030, 0.288]
hadd-greedy (privileged)	0.882 [0.774, 0.955]	0.928 [0.883, 0.966]	0.906 [0.883, 0.933]

Table 21: Per-seed D and three-seed D_3 (adapter M1 minus random-valid M1), and three-seed separation S (adapter M1 minus exact- ϵ rung M1), with paired task-clustered 95% intervals (10,000 draws, seed 133). Pre-registered entries are marked pre-reg. and the rest are descriptive.

	Validation (12)	P2 (11)	P2u (12)	Pooled (35)
D , seed 17	+0.328 [+0.155, +0.508]	+0.490 [+0.294, +0.688]	+0.196 [+0.031, +0.396]	+0.334 [+0.223, +0.450]
D , seed 29	+0.229 [+0.108, +0.354]	+0.268 [+0.093, +0.477]	+0.118 [+0.026, +0.220]	+0.203 [+0.124, +0.290]
D , seed 71	+0.297 [+0.134, +0.466]	+0.490 [+0.276, +0.694]	+0.133 [+0.031, +0.248]	+0.301 [+0.200, +0.411]
D_3	+0.285 [+0.140, +0.433]	+0.416 [+0.250, +0.591]	+0.149 [+0.030, +0.283]	+0.279 [+0.191, +0.375]
D_3 verdict	POSITIVE (pre-reg.)	POSITIVE (pre-reg.)	POSITIVE (pre-reg.)	POSITIVE (descriptive)
S vs 0.75	-0.041 [-0.187, +0.102]	+0.227 [+0.053, +0.403]	+0.002 [-0.145, +0.157]	+0.058 [-0.040, +0.156]
S verdict	NOT_SEPARATED (pre-reg.)	SEPARATED_ABOVE (pre-reg.)	NOT_SEPARATED (descriptive)	NOT_SEPARATED (descriptive)
S vs 0.50	-0.298 [-0.468, -0.129]	-0.075 [-0.247, +0.089]	-0.246 [-0.400, -0.082]	-
S verdict	SEPARATED_BELOW (descriptive)	NOT_SEPARATED (descriptive)	SEPARATED_BELOW (descriptive)	-

of its 10 screening episodes. The freeze walk, sorted by domain and seed, takes at most 2 tasks per domain up to 12 tasks, with a minimum of 8 tasks from 5 domains. The held-out panel froze with 11 tasks from 8 domains because only 11 admitted tasks were available under the per-domain limit, so it met the minimum and every held-out test is confirmatory.

The unscreened panel takes kept candidates outside the held-out panel on which the exact reference reaches the goal under both algorithms, without consulting the random-valid screen, at most 2 per domain, giving 12 tasks from 7 domains. All 12 unscreened tasks happen to have 0/10 random-valid goals in screening. All 638/638 held-out-panel and 696/696 unscreened-panel zoo episodes were independently replayed, with none missing.

The first freeze, from the original 40-seed pool 955000–955039, admitted 6 tasks from 5 domains and took the protocol’s blocked exit. Only controls-only screening had run, and no evaluation episode existed. An amendment then extended the seeds to 955000–955199. It kept the 480 earlier candidates unchanged and generated the new seeds 955040–955199 serially after them. The generation mechanism, the exclusion sources, the duplicate check, the C^* rule, R_t , the screen, the freeze walk, and the unscreened-panel rule stayed unchanged, and both panels were frozen from the combined pool before any evaluation episode.

Table 20 gives each arm’s M1 on the three panels. The four pre-registered adjacent gaps on the held-out panel are +0.130 [+0.070, +0.190], +0.239 [+0.183, +0.290], +0.302 [+0.259, +0.340], and +0.189 [+0.136, +0.248], so the ladder test passes. On the unscreened panel the descriptive gaps are +0.197 [+0.154, +0.243], +0.281 [+0.191, +0.355], +0.248 [+0.209, +0.290], and +0.147 [+0.088, +0.226], and every lower bound is positive.

Table 21 gives D per seed and D_3 with intervals on each panel and pooled over 35 tasks, and the separation S from the exact- ϵ rungs. Seed 29 has the lowest per-seed D on every panel, and on the held-out panel its separation point from the 0.75 rung is +0.080 (descriptive). No equivalence margin was pre-registered for S , so a non-separated S is not equivalence to the rung. The held-out separation rests on 11 task clusters, with 8 positive and 3 negative per-task differences, and percentile intervals can under-cover at this size.

Table 22 counts tasks by the sign of their three-seed difference and leaves out one domain at a time over the validation and held-out panels. Over the 35-task union 22 tasks are positive, 12 are zero, and 1 is negative. On the unscreened panel the eight zero tasks are both 15-puzzle, both blocksworld, both depot, and both driverlog tasks, on which neither the adapter nor random-valid solves within

Table 22: Concentration of the three-seed difference (descriptive). Top, tasks by the sign of the per-task difference. Bottom, D_3 over the validation panel and P2 with one domain left out, and the number of tasks left.

Panel	Positive	Zero	Negative
Validation (12)	9	2	1
P2 (11)	9	2	0
P2u (12)	4	8	0
Union (35)	22	12	1
Domain left out	D_3	95% interval	Tasks left
15-puzzle	+0.384	[+0.269, +0.501]	21
blocksworld	+0.395	[+0.269, +0.520]	19
depot	+0.353	[+0.224, +0.484]	20
elevators	+0.305	[+0.185, +0.430]	20
ferry	+0.331	[+0.212, +0.455]	21
grid	+0.331	[+0.204, +0.464]	20
storage	+0.332	[+0.209, +0.459]	21
Tower of Hanoi	+0.384	[+0.262, +0.507]	20
visitall	+0.318	[+0.211, +0.422]	22

$2R_t$. Every leave-one-domain-out D_3 lies from +0.305 to +0.395, and every lower bound is at least +0.185. All of these results are descriptive.

U.2 DAGGER ABLATION

This subsection reports the pre-registered DAGger ablation, for readers asking whether one on-policy round moves the adapter’s ladder position.

The pre-registered ablation collects on-policy records with the seed-17 adapter under each additive algorithm, aggregates them with replay records from the frozen corpus, retrains, and evaluates on the validation panel and the held-out panel. The primary endpoint is the paired separation S of the DAGger adapter from the exact- ϵ 0.75 rung pooled over the 23 tasks, with SEPARATED.ABOVE requiring a positive lower bound. The co-primary is the change Δ in M1 from the pre-DAGger adapter, with equivalence margin ± 0.05 .

An amendment at 11:21 UTC on 2026-09-24, decided before any DAGger training or evaluation outcome, cut the plan to seed 17 because the three-seed plan could not fit a hard wall-clock deadline. The running seed-29 collection was cancelled with 1.79 GPU-h still charged, and seed 71 was not run. Each algorithm’s training set became the first 512 on-policy seed-17 records plus 512 replay records, which the menu augmentation turns into 2048 samples over 64 steps, with the recipe otherwise unchanged.

All 24/24 validation-panel and 22/22 held-out-panel DAGger episodes were independently replayed with no mismatch. The primary endpoint is NOT_SEPARATED, $S = +0.115 [-0.026, +0.258]$ over 23 tasks, and the co-primary is INCONCLUSIVE, $\Delta = -0.030 [-0.114, +0.057]$. As descriptive secondary results, S is +0.278 [+0.082, +0.473] on the held-out panel and -0.034 [-0.197, +0.135] on the validation panel, and the last-label rate on the validation panel falls from 0.205 to 0.137. One small round at one seed therefore produces no detectable change in measured choice quality, since Δ is inconclusive against the ± 0.05 margin, and the ablation is inconclusive.

U.3 ZERO-SHOT POLICY

This subsection defines the zero-shot policy and reports its endpoints, for readers asking whether an untrained policy scores above random-valid.

The second policy is the pretrained Qwen3-VL-8B-Instruct base with no adapter, given the frozen scene-only choice-frontier observation and one appended system message. The message states the JSON format and asks for the frontier state whose scene appears closest to satisfying the goal pages. The policy uses a fixed three-rule output extractor, greedy decoding at inference seed 17, and the adapter’s cap of $2R_t$ calls. It ran on both additive algorithms over the validation panel and the held-out panel, 46 episodes in all.

The preferred arm, InternVL3.5-8B trained with the frozen three-seed recipe, was ruled out before any outcome. Its mean observation takes 11,748 tokens against 3,111 for Qwen3-VL, which puts training alone at 43 to 55 GPU-h against the 30 GPU-h cap. A reduced recipe would break the byte-

Table 23: Zero-shot base against random-valid (D_3), the exact- ϵ 0.75 rung (S), and the three-seed adapter (A), with paired task-clustered 95% intervals (10,000 draws, seed 133). Panel endpoints are pre-registered and pooled entries are descriptive. The p rows give exact sign-flip p values (descriptive). A p value tests a zero difference, and EQUIVALENT is the ± 0.05 margin rule.

	Validation (12, development)	P2 (11, confirmatory)	Pooled (23, descriptive)
D_3	+0.005 [−0.025, +0.044]	+0.001 [−0.048, +0.051]	+0.003 [−0.024, +0.033]
verdict	EQUIVALENT	INCONCLUSIVE	within margin (descriptive)
p	0.906	1.000	0.896
S vs 0.75	−0.321 [−0.422, −0.226]	−0.188 [−0.239, −0.140]	−0.257 [−0.324, −0.197]
verdict	SEPARATED.BELOW	SEPARATED.BELOW	SEPARATED.BELOW
p	4.88×10^{-4}	9.77×10^{-4}	2.38×10^{-7}
A vs adapter	−0.280 [−0.425, −0.142]	−0.415 [−0.589, −0.248]	−0.344 [−0.459, −0.232]
verdict	SEPARATED.BELOW	SEPARATED.BELOW	SEPARATED.BELOW
p	0.004	0.004	7.63×10^{-6}

identical condition, so it was not run in any form. The protocol was committed before any GPU launch, and no amendment followed.

Table 23 gives the endpoints. D_3 names the pre-registered learned-minus-random-valid endpoint for both arms (a single realisation for the zero-shot arm). Validation-panel verdicts are development-stage and held-out-panel verdicts are held-out confirmatory. On the held-out panel the upper bound of D_3 exceeds the ± 0.05 margin by 0.001, so the frozen rule returns INCONCLUSIVE rather than EQUIVALENT. The arm’s M1 is 0.026 [0.000, 0.063] on the validation panel, 0.017 [0.000, 0.051] on the held-out panel, and 0.022 [0.003, 0.046] pooled, below the exact- ϵ 0.75 rung on every panel. All 1,592 model calls produced a valid choice through the first extractor rule. Heap-head agreement minus chance is +0.003 and +0.001, the last-label rate is 0.328 and 0.285 against a chance rate of about 0.13 to 0.14, and 4 of 46 episodes solve at $2R_t$ (descriptive). The pre-registered prediction, D_3 within the margin with S and the adapter contrast A separated below on both panels, held. All 46/46 episodes were independently replayed with no mismatch.

V RESULTS DETAIL: EXTERNAL IDENTITY AUDIT

This appendix gives the interface survey and the ScienceWorld audit behind Section 5.4, with the LLM-First Search audit as a subsection.

The survey admitted published evaluations from 2022 on in which the model chooses from a runtime-supplied candidate set inside a planning or search loop, with public tasks and code. Seven candidates passed, five of type B (the model picks one valid action per step), two of type C (the model scores candidates that a search orders), and none of type A, our enumeration contract (Table 24). Four are not CPU-feasible under their papers’ budgets. For WebShop (free-text search, no step limit) and LLM-First Search (an unreported token cap) this was a judgement call, and LLM-First Search was the only saturated prediction, which the later audit in the LLM-First Search Audit appendix (Appendix V.1) tested. ScienceWorld is the most cited feasible candidate, and the runner-up shares its prediction. Six of the seven count budgets in steps, plan-length multiples, or iterations, and only LLM-First Search counts tokens. The closest published analogue, LFG (Shah et al., 2023), has the model score candidates in the enumeration style, and it was screened out because no evaluation harness was released.

The frozen protocol takes the first 200 test variations under the paper’s 100-step limit with five random-valid seeds (17, 5077, 6131, 7409, and 8527), and ten re-run episodes are byte-identical. The reference completes 183 of 200 (0.915), including 3/9 boil, 6/9 melt, 4/9 freeze, 6/9 change-state, 135/135 use-thermometer, and 29/29 measure-melting-point variations, and falls short only where 17 of the 36 Matter variations need more than 100 moves. Random-valid completes none, and 997 of its 1000 episodes end in a failure score.

The slice covers 6 of 30 task types (200 of 1,819 test variations), on all of which the paper’s own Random-Valid score is 0.00. The paper reports normalized scores, so its DRRN score of 0.084 on these tasks is not a success rate. The audit uses ScienceWorld 1.3.0, whose notes warn that gold paths can change relative to 1.2.x, and the paper’s simulator version was not re-run.

Table 24: The seven surveyed published interfaces and our two decision contracts. Type B: the model picks one valid action per step. Type C: the model scores candidates that a search orders. Type A: our enumeration contract. Prediction is the survey’s pre-audit prediction for random-valid against the reference. Audited gives the identity-audit result where an audit ran. Budget unit is the unit the source states.

Interface	Venue	Type	CPU-feasible	Prediction	Audited (identity result)	Budget unit
AutoPlanBench (Stein et al., 2025)	ICAPS 2025	B	yes	divergent	no	multiple of a reference plan length
ScienceWorld (Wang et al., 2022)	EMNLP 2022	B	yes	divergent	yes, 1000/1000 divergent	100-step limit
GLAM (BabyAI-Text)	ICML 2023	B	yes	divergent	no	environment steps (code)
WebShop	NeurIPS 2022	B	no	divergent	no	no step limit
RAP	EMNLP 2023	C	no	divergent	no	search iterations
LLM-First Search (Herr et al., 2025)	arXiv 2025	C	no [†]	saturated [†]	yes, no pair of the 400 identical	token caps
EmbodiedBench	ICML 2025	B	no	divergent	no	environment steps
Our enumeration contract (additive cells)	this paper	A	n/a	n/a	yes, 48/48 identical	decision calls
Our choice-frontier contract	this paper	n/a	n/a	n/a	yes, 18/18 divergent	decisions, twice the reference’s expansions

[†] Audited later with a GPU reference model, where saturation under the paper’s budget follows by construction (LLM-First Search Audit appendix, Appendix V.1).

V.1 LLM-FIRST SEARCH AUDIT

This subsection gives the LLM-First Search audit behind the budget check, for readers asking why a token budget cannot bind a chooser that spends no tokens.

LLM-First Search is a type-C interface. The model scores candidates, and the search expands the argmax child or pops the highest-valued frontier node, so every unexpanded alternative stays in the frontier. Random-valid answers every query with independent uniform draws, spends zero tokens, and makes no model call, and a decision is one node expansion after the root. The upstream code (commit 3025bdaa, Apache-2.0) is imported unchanged and not vendored. The fixed rule audits every configuration whose published GPT-4o win rate exceeds 10%, which excludes Sudoku 6×6 and keeps Countdown 3, 5, and 7 and Sudoku 4×4, with games 0 to 19 of each, 80 tasks in all. Budgets are the paper’s per-game token caps (1,000,000 for Countdown and 100,000 for Sudoku 4×4). Four deviations are declared. The paper’s GPT-4o is replaced by the local Qwen3-30B-A3B-Instruct-2507, so reference success is that model’s, the reference runs once per task, Sudoku 6×6 is excluded, and random-valid uses a faster frontier implementation that produced byte-identical logs on 57/57 probe episodes. Random-valid uses the five seeds of the first audit.

All 80/80 reference and 400/400 random-valid episodes were independently replayed with no mismatch. No pair of the 400 is identical, and the first divergence has median index 0 and maximum 2. Table 25 gives both readings. Under the paper’s budget random-valid wins every episode at every seed, so the verdict is SATURATED, and every reference loss ends at the token cap. This outcome follows by construction. Random-valid spends no tokens and the frontier keeps every alternative, so it is a complete randomized search, and a pre-audit probe had won 100/100 non-audited episodes. Its only way to lose was a 500,000-expansion safety guard, which never bound. Random-valid used a median of 201 expansions and a maximum of 102,606, against a median of 29.5 and a maximum of 1,300 for the reference. Within the reference’s own number of expansions per task, random-valid wins 0.0775 (per seed 0.0625, 0.075, 0.075, 0.0875, and 0.0875), so the verdict is CHOICE_REGISTERED. The reference’s matched-budget success equals its token-budget success by construction, so the budget curve in Table 25 is the evidence that choices matter. The matched-budget prediction held, and the paper-budget saturation follows by construction. Published GPT-4o success on these configurations is 0.768. One amendment, made before any audit episode, only reordered the smoke and audit start, and the ledger spent 2.89 of 12 GPU-h with no paid API use.

W RESULTS DETAIL: SEED REPLICATION AND FOLLOW-UP LIMITS

This appendix reports the enumeration-contract seed replication and lists the limits of every follow-up window, for readers checking seed and panel scope.

The prospective training-seed replication (seeds 17/29/71 frozen pre-launch, 534 new episodes independently replayed) executed on two cells (Table 5). In the greedy × multimodal replication cell (greedy best-first × multimodal-state, 24-problem unseen panel), per-seed success is 21/24, 21/24, 22/24 (mean 0.889, range [0.875, 0.917]). The contrast against the pretrained base is positive at every seed (+0.875/+0.875/+0.917, no sign flip). The contrast against random-valid stays negative (−0.125/−0.125/−0.083). On the DAgger iteration-1 cell set (BFS × three modalities, DAgger and continued-SFT arms), every arm and seed scores 0/72 unseen, with paired contrast +0.000

Table 25: LLM-First Search success by configuration under the paper’s token budget and at the reference’s own number of expansions. Differences are reference minus random-valid with task-cluster 95% intervals (10,000 draws, seed 133, descriptive). The reference is the substituted local model, and its matched-budget success equals its token-budget success by construction. Over all 80 tasks, random-valid success at 1, 1.25, 1.5, 1.75, and 2 times the reference’s expansions is 0.0775, 0.0975, 0.14, 0.17, and 0.19 (normalised area 0.135).

Configuration	Reference	Random-valid	Paper token budget	Random-valid	Matched decision budget	
			Difference		Difference	
Countdown 3 (20)	1.000	1.000	0.000 [0.000, 0.000]	0.120	+0.880	[+0.770, +0.970]
Countdown 5 (20)	0.750	1.000	−0.250 [−0.450, −0.100]	0.110	+0.640	[+0.370, +0.870]
Countdown 7 (20)	0.650	1.000	−0.350 [−0.550, −0.150]	0.020	+0.630	[+0.410, +0.840]
Sudoku 4×4 (20)	0.900	1.000	−0.100 [−0.250, 0.000]	0.060	+0.840	[+0.700, +0.950]
All (80)	0.825	1.000	−0.175 [−0.263, −0.100]	0.0775	+0.747	[+0.648, +0.838]
Verdict			SATURATED			CHOICE_REGISTERED

[+0.000, +0.000] at each seed. The published iteration-2 verdict (1/72 vs 1/72, seed 17) is unchanged. Both cells replicate across all three seeds. Cell A covers one of the three modalities of the greedy additive cell. The remaining two modalities and all other enumeration-contract cells use seed 17 only. Cross-seed aggregation is descriptive ($n=3$). The image-only-arm proposal has since executed under the native-arms v2 amendment (the Image-Only Native Arms appendix, Appendix Q), and its v1 gate remains as history.

The follow-up windows share this development-stage status, and we list their limits separately. The native-arms window evaluates an 18-unit clean panel with one training seed per cell. The R1-R4 follow-up window reuses the frozen native-arms adapters for R1, R2, and R4, and trains one seed-17 run per cell for R3. The choice-frontier window trains one seed per algorithm on a 9-task panel whose discriminative subset holds one task. The comparator zoo is a CPU-only exploratory analysis. The adapter has three training seeds on the validation panel, the held-out panel, and the unscreened panel, and the DAGger ablation trains one seed on half the planned corpus. None of these windows accessed the frozen 45-task final evaluation. The shared nine-task panel draws from 8 domains with 7 compact and 2 expanded variants.

Final primary evaluation prerequisites. No authorized protocol yet exists for the frozen 45-task final evaluation. Freezing one requires recorded decisions on the following prerequisites. (1) The condition-level data source and Search-Trace Segment construction with per-algorithm allocation. (2) Compute parity and checkpoint-selection rules. (3) The episode-budget unit, the whole-instance split procedure, and the fixed Search Memory capacity. (4) The estimand, the uncertainty interval or test, the success threshold, the invariant-compliance rule, and the validity and invalid-operation denominators for every cell. Until these choices are recorded, the final evaluation protocol cannot be frozen and no final comparative efficacy finding exists.

X DECISION CONTRACTS

Table 26 compares the enumeration and choice-frontier contracts defined in Section 3.

Y USE OF LLM STATEMENT

Large language model agents drafted and revised the manuscript text and wrote the figure scripts under the authors’ direction. An agent pass searched for citations, and a human verified every cited source. Agent reviewers ran repeated review rounds on the drafts. LLM agents also wrote and ran experiment and analysis code, drafted protocols and two pre-evaluation protocol amendments on the authors’ behalf, and compiled evidence sheets, and the authors approved each protocol before its outcome. The authors verified every reported number against its pinned artifact and take full responsibility for the content.

Table 26: The two decision contracts. Rows compare what the policy emits, who owns the frontier, what the observation carries, how the budget binds, and the role of the identity audit. Results under the two contracts are not comparable as performance rows.

	Enumeration contract	Choice-frontier contract
Policy emits	One Typed Search Operation per step, taken from a grounded, unscored candidate action menu	One opaque label from a menu of frontier states, naming the state to expand next
Frontier	Kept in textual Search Memory with visited and score state, advanced by submitted operations	Owned by the runtime. Menu labels are re-assigned per decision by a seeded permutation
Observation	Set by the Modality Observation adapter	Visual only, with task pages, the initial scene, one unlabelled 128px scene per frontier state, and goal pages
Budget	Learned arms capped at $2\times$ the exact-reference decision count, base at one call	Decisions and expansions capped at $2\times$ the additive exact-reference expansion count, goal selection uncharged
Identity audit	Diagnostic run on the expanded-baseline additive cells	Pre-registered pass condition over 18 control pairs